# ARC FLASH PLAYBOOK

Rev 01.1

Effective November 2025

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# Introduction to Arc Flash Safety

## Introduction to Arc Flash Safety

*Arc Flash Safety is critically important because arc flashes are among the most dangerous electrical hazards on data center projects. An **arc flash** is a sudden release of electrical energy through the air when a high-voltage gap exists and there is a breakdown between conductors.*

**WHY ARC FLASH SAFETY MATTERS**

Arc Flash incidents can severely impact people and projects in several ways.

1. **Severe Injuries or Fatalities**
   - Arc flashes can cause third-degree burns, blindness, hearing loss, and even death.
   - The blast can throw workers across rooms or cause them to fall from heights.
2. **High Medical and Legal Costs**
   - Treatment for arc flash injuries can cost hundreds of thousands of dollars.
   - Employers may face OSHA fines, lawsuits, and workers' compensation claims.
3. **Equipment Damage and Downtime**
   - Arc flashes can destroy electrical equipment, leading to costly repairs and production downtime.
4. **Regulatory Compliance**
   - OSHA and NFPA 70E require employers to assess arc flash hazards and implement safety measures.
   - Non-compliance can result in legal penalties and reputational damage.
5. **Workplace Safety Culture**
   - Promoting arc flash safety fosters a culture of awareness, training, and prevention, which benefits all aspects of workplace safety.

This Playbook outlines the responsibilities and workflows necessary to:

- Reduce the introduction of FOD (Foreign Objects and Debris) on job sites during construction
- Quality Control during equipment installation including FOK (First Of Kind) inspections
- Develop Energy Isolation Plans and establish/track the LOTO (Lock-Out, Tag-Out) Process
- CYT (Conditional Yellow Tag) process to safely provide temporary construction power to equipment for startup and testing

All trades, vendors, and project stakeholders have a role to play in preventing arc flash incidents. This playbook provides the training structure, inspection tools, and accountability systems necessary to achieve our safety and quality goals.

# Design

## Mitigation to Minimize an ARC Flash at the Design Stage

Mitigation should be considered in three separate categories:

**PASSIVE PROTECTION AGAINST ARC FLASH/ ARC BLAST-EQUIPMENT CONSTRUCTION**

Passive Protection against Arc Flash/ Arc Blast which involves switchgear being designed to mechanically withstand and contain /direct an Arc Flash or arc Blast by designing the mechanical protection into the construction and materials of the equipment or switchgear, they also utilise equipment to reduce the likelihood of an Arc Flash or arc Blast.

Equipment Design as an example, ArcBlok™ is designed with features that help avoid an arc flash before it happens. Barriers keep foreign objects, such as a hand, a dropped nut or screwdriver from entering the energized line side. Sensors inside the compartment continuously take thermal readings and communicate those to a mobile device, while maintenance personnel stand outside the arc flash zone to review. The system also self-extinguishes in less than one cycle to minimize equipment damage and injury to personnel


**ACTIVE PROTECTION AGAINST ARC FLASH / ARC BLAST-HIGH SPEED SENSORS**

Active Protection against Arc Flash / Arc Blast which involves equipment limiting the potential incident levels of an arc by its protective devices operating quickly and efficiently to reduce any potential incident level. Further to examples above High-Speed Light Sensing technology operates very efficiently with any change of light triggering a relay which opens any potential sources thus removing "fuel" for the arc or blast. Light sensors are positioned within equipment close to locations where current carrying conductors may be opening and closing.

**PREVENTATIVE PROTECTION AGAINST ARC FLASH/ ARC BLAST**

**Remote Switching**

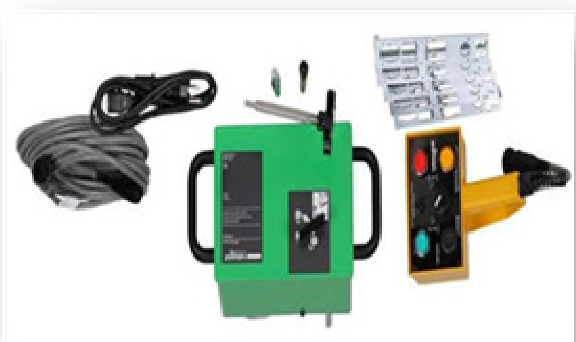
Preventative Protection against Arc Flash/ Arc Blast utilises remote operations such as remote open/ close and racking in/ out of circuit breakers. These can be designed to be a permanent solution in the network or can be employed temporarily as necessary. Maintenance regimes and procedures for areas and equipment should also be considered as preventative protection ensuring areas and equipment are debris free, insulators and insulation between current carrying conductors or current carrying conductors and earth have not been damaged or deteriorated and that all protective devices are operating correctly as per the associated protective settings.

### Remote Opening/ Closing

Remote switching, opening and closing of circuit breakers is a good method of protecting the operator from any potential incident. They can be installed permanently on MV and LV systems or if necessary, can be fitted temporarily although the circuit breakers on the project must be compatible. It is important to ensure that the operator, the ASP / Buddy and all personnel have vacated the area where you are likely to sustain an injury from an Arc Flash or Arc Blast and permanent solutions must consider the position of the remote operating location ensuring the operator is at a safe distance/ location.

### Remote Racking In/ Out

Racking in or out of circuit breakers is sometimes overlooked and greater consideration is given to the opening and closing of the device however, when racking a circuit breaker you are also at risk of causing an arc flash incident as there is



physical moving parts which may cause inadvertent movement or if operating in a dangerous environment, by positioning a conductor close to an exposed live conductor an arc may occur. Remote racking of circuit is often possible and can be implemented as part of the permanent design or implemented temporarily during switching activities/ high risk activities. When selecting protective devices thought should be given to the facility for remote racking in and out.


### IR Windows

Consideration of IR windows to be reintroduced into the design to aid with the thermal imaging testing/checks to be carried out.

Consideration of installation of temporary vent covers on the internal or external of the switchgear. This would eliminate further the chance of Foreign Object Debris making its way into the equipment during L2 site works.

### Arc Flash Study

Arc Flash studies calculate incident energy, considering factors like working distance, voltage, fault current, and upstream device clearing time. This data determines the required personal protective equipment (PPE) and the arc flash boundary, helping facilities protect employees and mitigate risks.

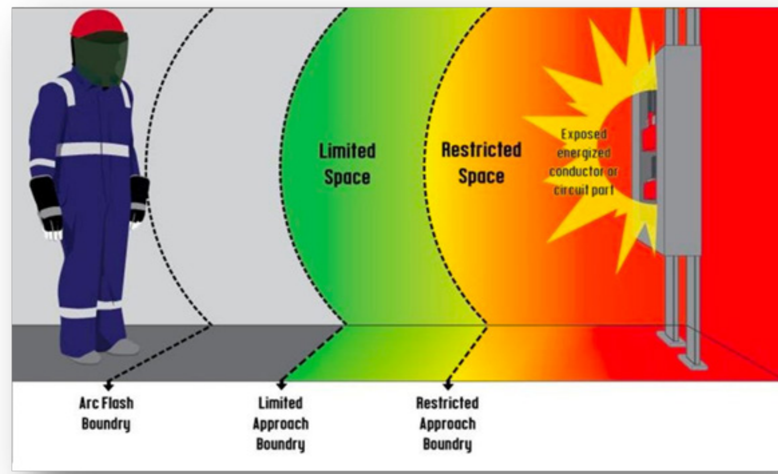

In order to complete an arc flash study, you must first model the entire power system using software such as SKM, CYME, Easypower, or ETAP.  A coordination study generally refers to a broad power system study – which includes short circuit analysis, protective device coordination and arc flash incident energy analysis.

An Arc Flash Study highlights several pieces of vital information to increase the safety of personnel operating or working in the vicinity of potentially energised equipment including:

- Equipment ID and Protective Device.
- Potential incident levels.
- Working Boundaries.
- Approach Boundaries.
- System Configuration or Scenarios.
- Arc Flash Warnings

Often an Arc Flash Study will highlight potential incident levels greater than 40 cal/ cm2 and suggest mitigation measures to be employed.

- Equipment ID and Protective Device- highlights and ensures the identification of the equipment and its source when the maximum potential incident levels are available.
- Potential incident levels- defines the maximum potential incident levels however, it also offers information on lesser potential incident levels when the system is configured differently with the equipment supported from a different source if available.
- Working Boundaries- defines the distance from the potential incident to the workers torso as all worst-case incident levels are defined with equipment open and no mechanical protection offered to the effected personnel.
- Approach Boundaries- defines approach boundaries as per the diagram:



**Arc Flash Boundary** - A distance from a prospective arc source at which the incident energy is calculated to be 1.2 Cal/ cm2. Experts have determined that the onset of second-degree burns occurs at 1.2 Cal/ cm2.

**Limited Approach Boundary** - An approach limit at a distance from an exposed energised electrical conductor or circuit part within which a shock hazard exists.

**Restricted Approach Boundary** - An approach limit at a distance from an exposed energised electrical conductor or circuit part within which there is an increased likelihood of electric shock, due to electrical arc-over combined with inadvertent movement.

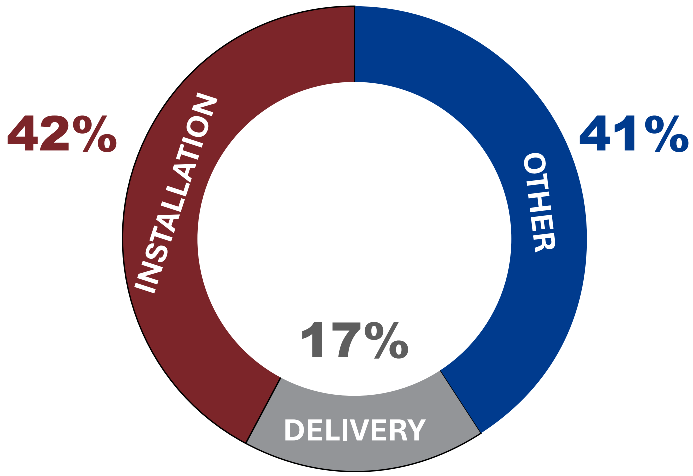
# QC/FOD
# *(Foreign Objects and Debris)*

## Introduction

*Foreign Object Debris (FOD) refers to any material, tool, or substance that does not belong in a specific location and poses a risk to people, equipment, or operations. In the context of electrical construction, FOD includes loose hardware, packaging remnants, metal shavings, dust, dropped tools, and personal items inadvertently left behind. While electricians may be the most directly affected by FOD, all trades contribute to and are responsible for its prevention. FOD is not just a cleanliness issue, it is a critical safety and operational concern.*


*In data center environments, the consequences of FOD are serious. Left unchecked, FOD can cause shorts, ground faults, arc flashes, equipment damage, and costly rework or downtime. Even seemingly minor debris—such as a screw dropped inside electrical gear or a tag left on a busbar—can trigger major electrical failures. As construction becomes more complex and just-in-time, the need for proactive and layered FOD prevention is essential.*

**ARC FLASH INCIDENTS BY PROJECT PHASE**




**WHY IT MATTERS**

Field data shows that 17% of arc flash incidents occur during the delivery phase and 42% during installation—most tied to preventable FOD conditions. These figures highlight the need for everyone onsite to adopt a zero-tolerance mindset around debris, organization, and gear protection.

**PARTS**

Any component, assembly, or other item that is installed in the product and intended to be a part of its configuration. Includes excess parts, spare parts, and test parts.

**CONSUMABLES**

Items that are used in the manufacture of the product but are not intended to be a part of the product's final configuration.

**TOOLS**

Items that are used to manufacture/test the product such as wrenches, screwdrivers, gauges, mirrors or any piece of one of these items.

**PERSONAL ITEMS**

Any item that originates from employees that is not normally a part of the production process. Examples include badges, pins, rings, and glasses.

**GENERAL DEBRIS**

True FOD. Any item, particle, or scrap made of any material that represents a potential hazard to the hardware or equipment such as metal shavings, dust/dirt, and strands/fibers of thread or wire.

## SOURCES OF FOD

The most common sources of FOD on Data Center Projects:

- Nuts, screws and bolts
- Tools or materials dropped or left in gear
- Dust from work and/or rework (metal shavings, drywall dust, fireproofing, insulation, painting and not recleaning after work)
- Packaging not fully removed and external filters in place during construction
- Excessive dust in air (exterior conditions or interior cleaning methods)
- Rodents and other animals
- Construction waste and debris that is left behind
- Testing equipment or loose cable/loom left after testing

## PREVENTION STRATEGY

- Designating FOD-sensitive zones (Awareness, Control, and Critical) and assigning appropriate levels of access and oversight
- Conducting regular FOD walkdowns and formal inspections before energization
- Requiring "clean-as-you-go" practices for all trades, enforced daily by FOD Captains
- Establishing robust training, tool control, and tamper seal protocols
- Mandating layered signoffs—especially during First-of-Kind (FOK ) installation and inspections, Pre-energization inspections (L2 or Yellow Tag) , and Pre-Energization processes

FOD prevention isn't one person's job. It's a shared responsibility that directly impacts site safety, schedule certainty, and system performance. This playbook defines the standards, responsibilities, and best practices that protect people and critical infrastructure from hidden hazards.

# FOD Training

Training equips personnel with the knowledge to prevent, detect, and remediate FOD hazards effectively.

Training Elements:

- **Explanation of FOD:** What it is, how it's introduced, and its impact on systems
- **Zone identification:** Awareness, Control, Critical - each with its own level of precautions and access requirements
- **Source identification:** Understanding how normal activities (cutting, packaging removal, tool use) contribute to FOD
- **Mitigation:** Clean-as-you-go, 6S housekeeping, protective barriers
- **Accountability:** Logging tools and materials, ensuring nothing is left behind
- **Escalation:** How and when to report FOD

### Training Schedule:

- Conducted at site induction and reinforced:
- At phase transitions
- After major equipment deliveries
- Prior to energization
- Following FOD-related incidents or near-misses

**TOOLBOX TALK - ALL TRADES**

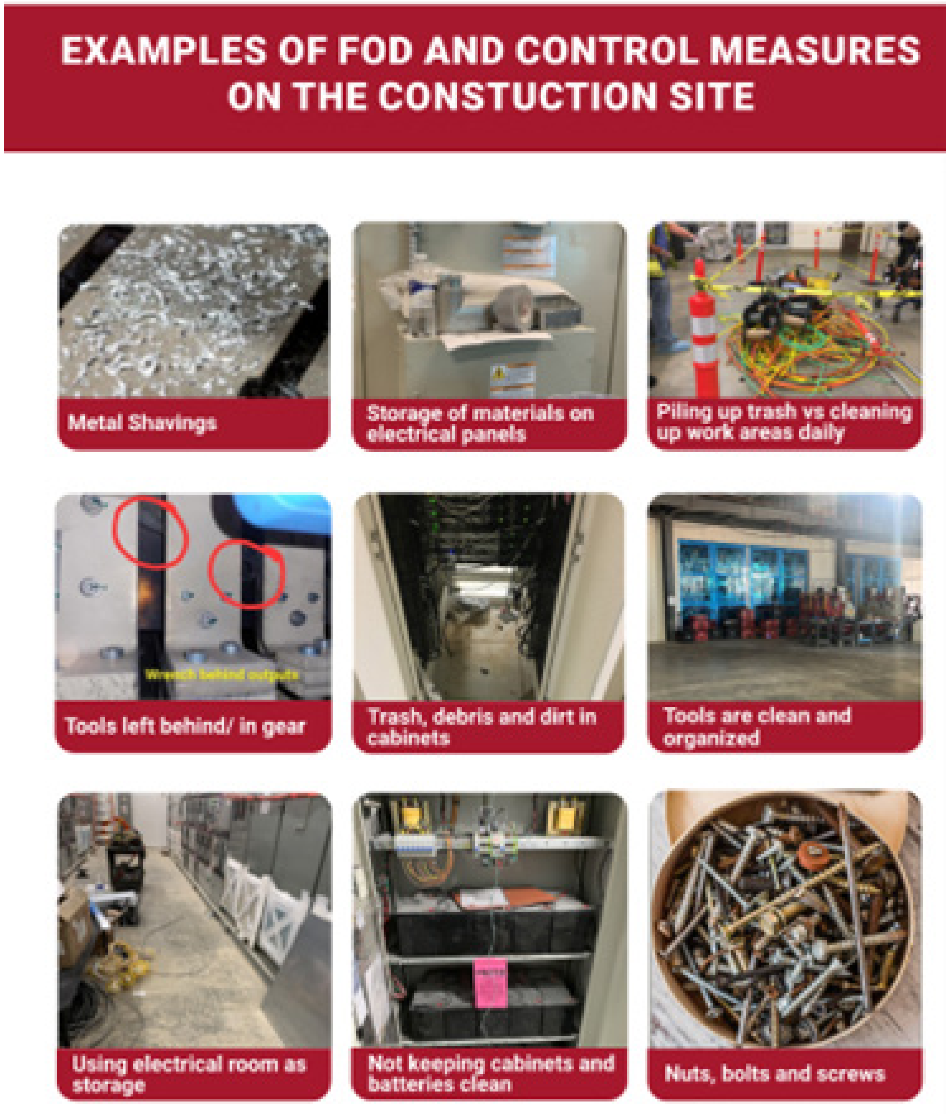
Toolbox talks reinforce safety messaging in a quick, practical format. This one ensures all site personnel understand how their daily work affects FOD risk. It includes the following topics:

### Discussion Points:

- Define FOD and cite real examples (e.g., a bolt left in a panel)
- Discuss trade-specific FOD risks and how each trade contributes
- Introduce practical steps: zip tie loose items, label and bag fasteners, maintain dedicated tool storage
- Emphasize that FOD isn't just an electrical issue , reference sources of FOD page 7.
- Highlight consequences: potential injury, rework, delays, and liability

Reiterate the importance of shared accountability in maintaining a clean, safe site.


*Image: FOD Awareness Summary and Examples from Toolbox Talk*




**TOOLBOX TALK - STRUCTURAL / DRYWALL**

This trade-specific FOD awareness and prevention talk targets drywall installers, ceiling grid crews, and related structural workers who often generate airborne dust and fine debris. These materials are among the highest risk factors for arc flash-related contamination.

Toolbox Talk content should include:

- **Drywall Dust Risk Education**: Clearly outline the threat posed by drywall dust, gypsum particulates, and finishing compounds. Even trace levels of dust can penetrate enclosures, causing electrical tracking and thermal failure after energization.
- **Saw and Tool Restrictions**: Reinforce that mechanical saws and rotary tools are not permitted within or near electrical rooms. Cutting must be moved to designated cut stations outside FOD-sensitive zones.
- **Dust Containment Measures**: Require the use of enclosed cut stations with active dust collection or vacuum attachment systems with suitable dust filtration materials. Teams must utilize portable negative air units or vented enclosures when sanding near energized or soon-to-be-energized areas.
- **Daily Cleaning Protocols**: Structural trades must complete broom sweeps multiple times per shift and using vacuums with suitable dust filtration materials in and around electrical corridors, electrical Rooms, mechanical rooms, Data Halls, and other spaces with electrical equipment. Tools and materials should be kept off the floor and out of electrical pathways.

- **Work Surface Protection**: Encourage use of magnetic drop cloths, ceiling wrap protection, or temporary adhesive mats directly below areas of active overhead work. These tools help catch falling dust and particles before they reach sensitive equipment.

This training should be mandatory for all drywall crews operating within 50 feet of any designated Control or Critical FOD zone and must be refreshed monthly, or prior to any Yellow or Blue Tag milestone.

**TOOLBOX TALK - MEP**

This training targets mechanical, electrical, and plumbing trades involved in installation, tie-ins, and system modification around energized or soon-to-be-energized gear. MEP (Mechanical, Electrical, Plumbing) scopes often present heightened FOD risks due to invasive work on or around equipment.

Toolbox Talk content should include:

- **Cutting Prohibitions**: Cutting of Unistrut, conduit, or pipe is strictly prohibited inside any electrical room or FOD Critical Area. All mechanical cutting operations must be relocated to designated external cut zones with dust collection measures.
- **Tool Control at Elevation**: All trades working above gear must use tool lanyards or tethered tools. Unsecured hand tools are a primary source of dropped-object FOD incidents.
- **Temporary Protection During Penetrations**: When drilling, tapping, or modifying panels or enclosures, temporary protection such as magnetic covers, flexible shields, or containment sheeting must be applied to prevent debris intrusion.
- **Pipe Termination Safety**: All exposed or incomplete pipe ends must be sealed with approved temporary caps or plugs. Foam, tape, or makeshift covers are not acceptable.
- **Post-Installation Cleanup Standards**: Upon completion of installation or tie-in activities, interior cabinet and enclosure spaces must be vacuumed with suitable dust filtration materials . Any work generating shavings, insulation, or residue requires post-task inspection by the GC QC or Energy Marshal prior to panel closure.

Training for MEP trades should be completed during onboarding and reissued before commissioning phases. Compliance should be confirmed during FOK and L2 inspections .

**FOD INSPECTION TRAINING**

Inspection is a cornerstone of FOD prevention, and qualified inspectors must be equipped to recognize FOD in all forms.

Training includes walkthroughs of:

- NFPA 70E : Electrical safety awareness and PPE use
- LOTO: Specific lockout/tagout sequences for energized and near-energized areas
- Use of checklists and inspection forms tailored to equipment and room type
- Recognition of conductive vs. non-conductive contaminants
- Best practices for thorough inspections: lighting angles, mirrors, vacuums, and swabs
- Documentation and photographic evidence for quality tracking

***See Exhibit A for a sample inspection form.***

## FOD TRAINING TIMELINE

- **Day 1 (Onboarding):** All trades - FOD Toolbox Talk included with Safety Orientation
- **Monthly**: Reinforce through stretch-and-flex discussions
- **Before first Yellow Tag**: Re-fresh training and assign QC FOD Captains; to reinforce equipment cleaning measures inclusive of equipment O&M instructions.
- **Prior to IST (Integrated Systems Test )**: Re-certify cleaning staff and QC team

Training Milestones:

- **Site Mobilization**: Introduction to site-specific FOD rules
- **Trade Mobilization**: Tailored training for electrical, structural, and MEP crews
- **Pre-Commissioning**: Focused refreshers on cleaning, inspection, and reporting
- **Energization Prep**: Specialized sessions for Yellow Tag processes and tamper seal protocols
- **Post-Incident**: Corrective training following any FOD discovery or deviation

Training must be documented with attendance logs and trainer certifications.

# FOD Risk Areas

Effective FOD prevention starts with clearly defined risk zones, each requiring escalating levels of control based on proximity to electrical equipment and potential impact to energization integrity.

FOD risk zones are divided into three categories based on potential impact and required level of control. Each area should be clearly defined on the site map and physically demarcated with signage and barriers where appropriate. These zones adjust with construction and shall be updated as equipment or sensitive installations come into play.

### AWARENESS AREAS (GENERAL AWARENESS)

These are lower-risk zones where general work takes place but FOD still poses a threat. Ex. General circulation or staging zones that require baseline cleanliness and vigilance such as loading docks, cable pathways, and equipment corridors.

Requirements:

- No cutting or grinding activities in designated areas or in the vicinity of equipment
- No open material storage
- Emphasis on good housekeeping
- Tamper seals in place and logged prior to FTE

### CONTROL AREAS (FOD LIMITED ACCESS AREAS)

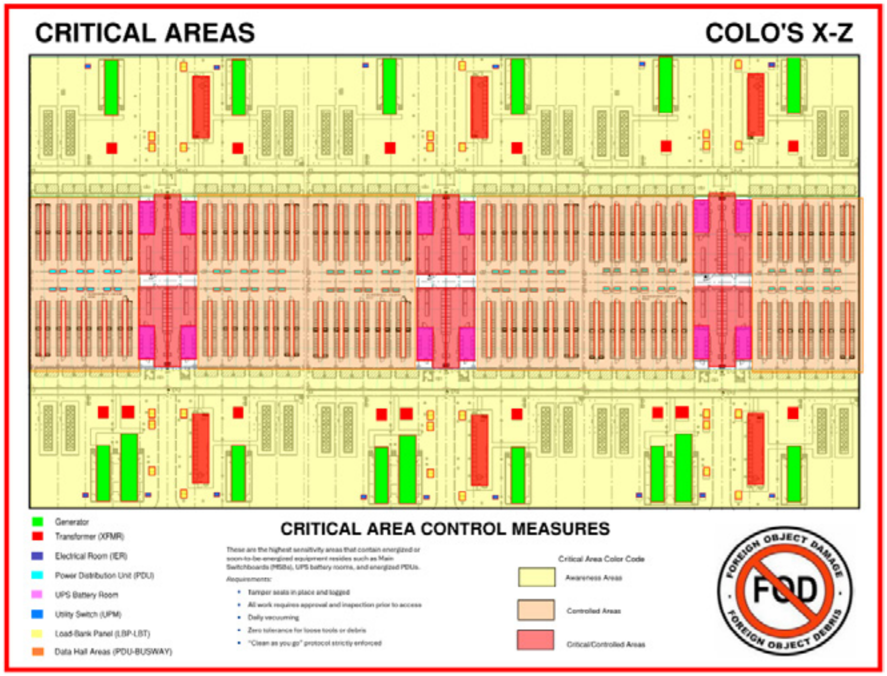
These are workspaces where installation or modification of equipment occurs or sensitive activities are ongoing such as panel prep zones, gear modification areas, and unsealed electrical equipment rooms.

Requirements:

- Restricted access (sign-in required)
- No cutting or grinding activities
- Post-task cleanup and inspection mandatory
- Use of covers and containment measures during intrusive work
- Tamper seals in place and logged prior to FTE

### CRITICAL AREAS (CRITICAL ACCESS CONTROLLED AREAS)

These are the highest sensitivity areas that contain energized or soon-to-be-energized equipment resides such as Main Switchboards (MSBs), Uninterrupted Power Supply (UPS) battery rooms, and energized Power Distribution Unites (PDUs ).



Requirements:

- Restricted access (sign-in required)
- Tamper seals in place and logged prior to FTE
- All work requires approval and inspection prior to access
- Daily vacuuming with HEPA units
- Zero tolerance for loose tools or debris
- Tool tracking required for work on previously energized equipment
- “Clean as you go” protocol strictly enforced

**VISUAL TOOLS FOR FOD RISK AREA IDENTIFICATION**

Visual management is critical to the success of a FOD prevention program. Color-coded maps, standardized signage, and consistent field communication help all trade workers recognize and respect zone-specific controls - especially in fast-moving, high-risk environments like data centers.

### Color-Coded Area Maps

Incorporate clearly marked FOD prevention zones directly into construction floor plans, posted at the following locations:

- Site entrances
- Shift huddle locations
- Electrical rooms and corridors
- Equipment staging and delivery areas

Use the following universal FOD color-coding scheme:

**Yellow = Awareness Areas (Green or Blue permitted where it exists until phased out**

Denotes general-use areas where equipment may be nearby but not actively worked on. Minimal risk, but cleanliness and vigilance are required.

**Orange = Control Areas**

Represents controlled-access zones where gear assembly or modification is ongoing. No cutting or grinding work permitted.

**Red = Critical Areas**

Highest-sensitivity environments. Gear in these areas is energized or pending energization. Access is restricted to qualified personnel with task-specific signoff. No cutting, grinding, or unsupervised work permitted.

*Tip: Overlay these zones in project BIM (Building Information Modeling ) models and export PDF versions for posting on site and inclusion in FOD Toolbox Talks.*

### Jobsite Signage

FOD area signage must be posted at all major entry points to designated zones. Signs should:

- Use high-contrast colors matching the zone classification (green/orange/red).
- Include simple icons (e.g., broom, shield, lightning bolt) to help communicate requirements quickly across language barriers.
- Clearly display area-specific instructions:
- “No tool storage”
- “HEPA vacuum required after work”
- “Authorized personnel only – energized equipment”

Reusable laminated signs should be printed in English and Spanish, with QR codes linking to the FOD Prevention Plan or Clean-As-You-Go checklist.

### Field Communication and Orientation

- During site orientation, all workers must be walked through the zone types using floor plans and sample signage.
- Pre-shift huddles should include FOD map reviews when work occurs near Critical Areas.
- Trade foremen / FOD Captains must carry zone maps in their field binders for reference during inspections and work planning.

By integrating these visual tools into daily routines, the project team reinforces a shared safety culture where every worker can recognize, respect, and act within the defined FOD control framework.

# FOD Prevention Plan

The FOD Prevention Plan is the foundation of any quality assurance effort aimed at reducing arc flash risk. It includes:

- **Zoning Designation:** Assign and map out Awareness, Control, and Critical areas for each phase of construction and commissioning. Post signage on-site.
- **Trade Coordination:** Establish responsibilities using a RACI matrix . The General Contractor, subcontractors, and vendors must have clarity on who owns what portion of prevention.
- **Housekeeping Protocols:** Implement 6S (Sort, Set in Order, Shine, Standardize, Sustain, Safety) and clean-as-you-go standards for every shift.
- **Debris Controls:** Metal shaving control, tool tracking, containment barriers, no cutting inside gear rooms, and proper material staging away from gear.
- **Barrier Protection:** Use magnetic trays, panel cover carts, and sealed equipment when not in active use.
- **Daily Execution:** Visual inspections at start, midpoint, and end of shifts. Require photo documentation before closing equipment.
- **Tool Control Systems:** Track issued tools per person. Require sign-offs and inspections prior to panel closure.

***See Exhibit B for a FOD Prevention Plan Template***

To prevent and control FOD, implement the following practices:

- Provide FOD awareness training to all personnel
- Pave around site as soon as possible and use a water truck and street sweeper to keep dust to a minimum. Have boot cleaners by door and walk off mats inside to collect dirt before spreading it throughout building. Use boot coverings after energization of spaces
- Conduct regular FOD walks and inspections where on the spot corrections occur
- Use tool control systems and checklists after yellow tagging equipment
- Maintain clean and organized workspaces
- Daily housekeeping and limit broom sweeping. Use floor scrubbers and/or vacuums with appropriate filters media.
    - Limit the amount of ‘dirty’ outside tools that are used inside (especially after energization)
    - After energization has started, cut materials outside and bring into the data halls or space to minimize the loose particles or shavings
    - Do not lean tools or materials on gear or store them in electrical rooms
    - No storage of materials in, on top of, or around electrical gear
    - Leave the temporary protection on electrical gear as long as possible
- Sequence activities to minimize work above the gear
    - Tool tethers shall be used when working over gear to limit dropped objects
    - Gear and vents shall be covered if work is occurring above it

    - Covers shall be kept on gear and reinstalled daily or after work concludes
- Secure protective covers and use containment for loose items
    - Utilize magnetic trays for nuts and bolts
    - Reinstall screws/bolts to equipment after removing covers so they are not lost
    - Use panel cover carts or a designated storage cart to securely store panels
    - Ensure all holes/knockouts are enclosed with grommets and wires or covers
- Report and remove FOD immediately upon discovery

## FOD PREVENTION BEST PRACTICES

### Leadership and Culture

- Assign FOD responsibility to team leads to establish clear accountability.
- Promote a culture of care and ownership among all team members.
- Regular FOD awareness training for all personnel, including toolbox talks and stretch-and-flex meetings.

### Daily and Scheduled Inspections:

- Implement a daily FOD prevention checklist to maintain consistent standards and accountability.
- Conduct daily FOD inspections and pre-energization FOD inspections at any opened gear.
- Utilize First-of-Kind (FOK) inspections as mock-ups prior to critical tasks.
- Use room readiness checklists before energization.
- Maintain detailed inspection logs and report any FOD findings immediately.

### Signage and Communication:

- Clearly mark FOD-sensitive areas with signage to reinforce the importance of cleanliness and control.
- Post visual reminders and guidelines prominently throughout the site.

### Equipment and Material Handling:

- Request to have equipment shipped on plastic pallets instead of wood as much as feasible to reduce risk of splinter debris .
- Protect electrical gear after delivery using physical barriers such as K-rails, bumper blocks, plastic chains, and Clearline rubber blocks to prevent accidental damage.
- Utilize panel cover carts and organized tool carts for managing equipment covers and tools securely.
- Store tools, parts, and materials in designated, labeled storage areas away from electrical equipment.

### Tool and Material Control:

- Use tool tethering when working above equipment to prevent dropped objects.
- After equipment energization, tools will be accounted for at the completion of any MOP conducted, accessed gear shall be re-inspected and resealed. Implement magnetic trays and secure all loose items immediately.
- Reinstate equipment covers daily and reinstall screws/bolts immediately after removal. Utilize temporary protection when conductors or other ongoing work prevent permanent cover from being applied.

### Metal Shavings and Dust Prevention:

- Perform cutting operations outside the building when possible, using snipping tools instead of cutting tools to reduce shavings.
- Establish dedicated "cut zones" and use enclosed stations if inside cutting is unavoidable.
- Use approved containment techniques to prevent dust migration, especially in energized data center environments.

### Protective Covers and Physical Barriers:

- Keep temporary protection on electrical gear in place as long as possible.
- Ensure all vents, openings, and sensitive areas are sealed with temporary covers.
- Utilize protective covers and barriers during any overhead or adjacent work.

### Housekeeping Practices:

- Maintain organized workspaces; implement daily housekeeping and "Clean as You Go" strategies.
- Employ continuous debris removal during fabrication, installation, or maintenance tasks.
- Schedule dedicated cleaning times, such as end-of-shift clean-ups.
- Use proper cleaning tools, such as HEPA-filtered vacuums, and minimize broom sweeping.
- Limit the use of "dirty" outside tools within controlled areas, especially post-energization.

### Fire Prevention and Safety:

- Replace dry chemical fire extinguishers with Halotron fire extinguishers in energized data hall environments to reduce residue risks.
- Ensure permanent fire extinguishers and emergency equipment are in place before energization.

### Environmental Controls:

- Implement measures for rodent and environmental contamination prevention early in the project.
- Use walk-off mats and boot cleaners at building entrances, complemented by protective boot coverings post-energization.

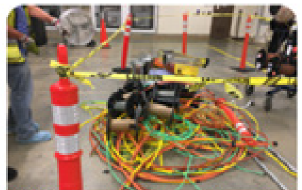
### Documentation and Reporting:

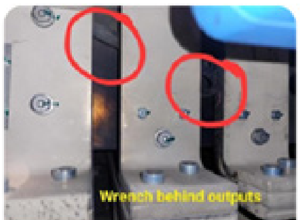
- Establish clear reporting mechanisms for incidents and corrective actions.
- Document all FOD-related processes, procedures, and inspections thoroughly.

## EXAMPLES OF FOD AND CONTROL MEASURES

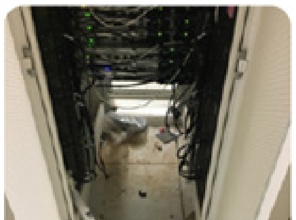

Metal Shavings

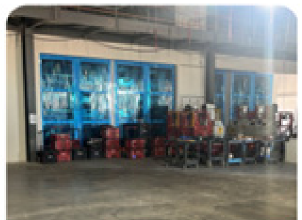

Storage of materials on electrical panels

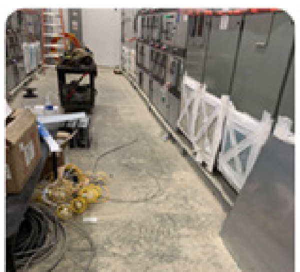

Piling up trash vs cleaning up work areas daily

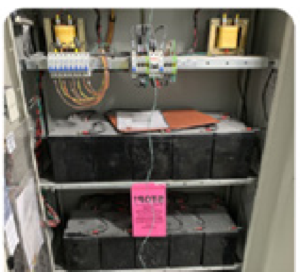


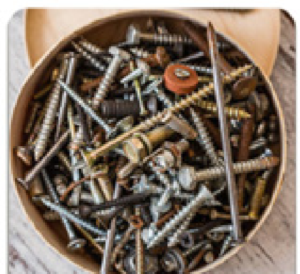
Tools left behind/ in gear


Trash, debris and dirt in cabinets


Tools are clean and organized


Using electrical room as storage


Not keeping cabinets and batteries clean


Nuts, bolts and screws

# FOD Reporting Process

All personnel - regardless of trade, role, or rank - are both empowered and obligated to report any instances of Foreign Object Debris. Quick response and thorough documentation are essential to prevent unsafe conditions that could result in arc flash events, equipment damage, or schedule delays.

### Reporting Tools and Actions:

- **Daily FOD Inspections :** All crews inside the building,  generators, or power centers must complete and document Daily FOD Inspections at opened gear  to verify the work zone and equipment are clear of debris. Suggested Checklist Templates are provided in the appendix
- **Post-Yellow Tag  (Level 2) Discovery:** If any FOD is discovered after equipment has been Yellow Tagged, stop work immediately and alert the GC Quality Control lead and the Energy Marshal. No further action can occur until a full inspection and remediation is complete.
- **Root Cause Analysis (RCA):**  For FOD found inside any equipment, post tamper seal a measured  RCA must be conducted. This includes capturing photographic evidence, logging the date and time, identifying the last personnel to work on the equipment, and assessing the potential impact on operations and safety.
- **Pre-Energization Verification:** Before energizing any equipment, a complete inspection and signoff must occur using the Pre-Energization FOD Inspection Template.
- **Near Miss Reporting:** Any situation where FOD had the potential to cause harm - especially one that could have led to an arc flash - must be logged as a near miss and submitted to the site safety team for review and learning .
- **FOD Program Audits:** Site safety team to audit overall FOD Program for implementation and effectiveness. Documentation can be aligned with other project specific safety audits.

FOD discovered after Yellow Tag issuance is considered a critical event. Debris that cannot be recovered or located must trigger an incident report and Root Cause Analysis (RCA) . No gear may be energized until all items are accounted for, inspections are complete, and the condition is cleared by the Commissioning Authority (CxA)  and GC Quality Control (QC)  leadership.

# Tamper Seal Process

Tamper seals are a critical quality control mechanism and serve as physical confirmation that electrical equipment has passed all required inspections, been properly cleaned, closed, and is ready for safe energization. Their presence signifies that no further intrusive work may occur without formal approval, inspection, and re-sealing.

### Purpose of Tamper Seals

- Prevent unauthorized or undocumented rework inside gear after it is deemed ready.
- Reduce risk of arc flash, short circuits, or foreign object intrusion by ensuring sealed conditions remain intact.
- Create an auditable trail of inspections and access points from panel closure through to turnover.

**TAMPER SEAL APPLICATION PROTOCOL**

Tamper seals must only be applied after a series of quality checks are completed, and all work inside the enclosure is finished.

Steps:

1. **Perform final FOD inspection** using the Pre-Energization Checklist.

   ***See Exhibit C – Pre-Energization FOD Inspection Template.***

2. **Reassemble the equipment panel(s)** - confirm all screws, gaskets, and covers are in place.
3. **Take photo documentation** of internal condition and panel closure (retain in QC logs).
4. **Apply tamper seal** across the main point of access - seal must not be removable without breakage.
5. **Record the seal's serial number** in the Tamper Seal Log (equipment ID, date, inspector name).
6. **Notify CxA and Energy Marshal** of seal application for readiness validation.

**BREAKING TAMPER SEALS**

Tamper seals must only be broken under specific, pre-approved conditions (i.e MOP's, Permits, Scripts), and the breakage must immediately trigger a defined inspection protocol.

Approved Reasons to Break a Seal:

- Execution of a formal LOTO procedure
- Pre-scheduled IR scanning
- Directed CxA commissioning test with onsite supervision

If a Seal is Broken (Approved or Not):

- All work must stop in that zone
- Notify the GC Superintendent, CxA, and Energy Marshal
- Perform a full FOD inspection
- Capture photos of interior condition
- Reapply new tamper seal and log serial number
- Include a note in the Daily QC Report and update inspection logs

**PROHIBITION OF POST-SEAL WORK**

Once a tamper seal has been applied:

- **No further intrusive work may occur**
- No minor adjustments, terminations, or label additions are permitted
- Entry requires full re-authorization, resealing, and documentation
- Any attempt to bypass or ignore seal protocol is subject to disciplinary action and site escalation

**CXPOR 8.2 ENCLOSURE E - TAMPER SEAL PROCESS**

Seal integrity is a contractual requirement and quality control gate. The Microsoft Commissioning Plan of Record (CxPoR) document provides the definitive procedure for tamper seal use across all project phases:

| PHASE | RESPONSIBILITY | TRIGGER/REQUIREMENT | PROCEDURE |
|---|---|---|---|
| **Construction** | GC owns the Tamper Seal Log and applies seals | Seals applied after panel closure and Yellow Tag milestone | If seal is broken, GC must reinspect and re-seal the equipment |
| **Acceptance (after L3 Cx complete)** | CxA assumes ownership of the Tamper Seal Log | Blue Tag application requires all seals verified intact or replaced under CxA supervision | CxA validates seal condition before energization or commissioning |
| **Turnover** | Shared responsibility (CxA/CxM for inspection and signoff) | White Tags cannot be applied to gear with broken, missing, or undocumented seals | Final seal condition must be inspected and documented in turnover acceptance package |

*Diagrams from CXPOR8.2 Enclosure E should be printed and posted at each gear room and commissioning trailer to reinforce process awareness.*

**BROKEN TAMPER SEAL TRACKING**

Tamper seal breaches - intentional or not - must be logged, investigated, and reported. Broken seals must be treated as critical incidents:

### Log Data Points:

- Date, time, equipment ID
- Person who discovered break
- Description of condition upon discovery
- Whether break was approved or unauthorized
- Responsible party (if known)
- RCA documentation

### Corrective Actions:

- Immediate FOD inspection and cleanup
- CxA or GC reinspection of gear interior
- Photo documentation of findings
- New tamper seal applied and logged
- Notification to commissioning team and Energy Marshal

### Trend Monitoring:

- Weekly reviews of the Broken Seal Log by QA/QC leadership
- Patterns of frequent breakage may indicate:
- Inadequate work planning
- Improper task sequencing
- Lack of training or supervision

Where trends are identified, GC and Energy Marshal develop a Corrective Action Plan (CAP) and deliver it to the CxA and Microsoft PM.

**TAMPER SEAL AUDIT**

A formal tamper seal audit validates the integrity of the tamper seal process and serves as a final check to ensure equipment is secure, unaltered, and ready for energization. This process reinforces procedural discipline, identifies documentation gaps, and strengthens project closeout.

### Audit Timing:

- End of Construction Phase (before transition to commissioning)
- Pre-Yellow Tag issuance (per system or zone)
- Prior to Blue Tag (CxA acceptance milestone)
- Before White Tag and final Turnover

### Audit Procedure:

- Conduct physical walkthroughs to confirm presence and condition of all tamper seals on tagged equipment
- Cross-reference each seal's serial number with entries in the official Tamper Seal Log
- Investigate any discrepancies: missing seals, mismatched serials, or undocumented breaks
- Interview relevant trade workers or supervisors if unexplained activity is suspected
- Confirm that all previously broken seals were properly logged and reinspected

### Reporting:

- Compile findings in a formal QC Memo detailing:
    - Total number of seals reviewed
    - Seal serial numbers, equipment IDs, and locations
    - Any missing, damaged, or non-compliant seals

    - Observations or interviews conducted

- Submit the report to the Commissioning Manager and Microsoft Quality Assurance (QA) for validation
- Log audit results in the Automated Commissioning Management System (ACMS) or project binder as a closeout deliverable
    - *Best Practice Tip: Use photos to document seal conditions and affix a sticker to the enclosure noting audit date and initials of the auditor.*
- Assign follow-up actions for any unresolved issues

# Quality Control

The Quality Control team holds final authority for validating system readiness prior to energization. QC is empowered to stop work, inspect for compliance, and require remediation if conditions do not meet the project's FOD and safety standards.

**Key Responsibilities:**

- **FOD Sign-Off:** No equipment enclosure may be sealed until QC visually and photographically verifies:
  - Internal cleanliness and absence of FOD
  - Correct installation according to shop drawings and submittals
  - Proper labeling (gear, breakers, terminations, grounding, etc.)
  - Fastener torque as per manufacturer's recommendations
- **Tool Management:** Enforce tool check-in/check-out procedures in high-risk zones (UPS rooms, MSBs, etc.) using physical logs or digital inventory systems. Verify nothing is left inside gear post-task.
- **Photo Documentation:** All inspections must be supported with timestamped photos stored on the project database or shared server, labeled by equipment ID and location.
- **Push Process Oversight:** QC initiates and monitors any deviations through the Microsoft Push Log. All issues requiring exception from energization protocols must receive written approval prior to continuation.

The QC team actively engages at multiple points in the project to uphold safety and compliance. They participate in First-of-Kind (FOK) inspections alongside trade partners and vendors to ensure early-stage quality benchmarks are met. During readiness walkdowns and pre-functional checkouts, QC verifies that installation standards, FOD cleanliness, and documentation align with requirements. Before any Yellow Tag is issued, QC serves as the final checkpoint, confirming gear readiness and procedural adherence. Throughout the project, they maintain a comprehensive audit trail of FOD and inspection activities that form part of the final turnover package.

## ROOM READINESS CHECKLIST

Room readiness ensures that the entire environment—not just the gear—is safe, clean, and properly labeled before electrical systems are energized. This process mitigates downstream risks of contamination, unauthorized access, and inspection failure.

**Checklist Requirements:**

- **Cleanliness:** All surfaces must be free of dust, trash, and construction debris. Floors swept or vacuumed; overhead areas visually inspected.
- **Equipment Labels:** Each cabinet, PDU, ATS , and switchboard and like electrical equipment must be labeled with permanent markers per the one-line diagram.
- **Panel Integrity:** All covers must be secured and fastened. Tamper seals should be in place and recorded in logs.
- **Signage and Boundary Marking:**
  - Room number and permit sleeve at entry
  - "Danger – Energized Equipment" signage
  - Arc flash warning labels installed on gear
  - Clear Controlled Access Zone (CAZ) boundary lines on floor

- **Verification Walkdown:** Final inspection is conducted by GC, Commissioning Agent (CxA), and client representative, with a signed checklist retained in the turnover package.

***See Exhibit D Room Readiness Checklist for Energization***

**EQUIPMENT PROTECTION PLAN**

All electrical gear should be protected from physical damage, environmental contamination, and unauthorized access during all stages—from offloading and staging to final commissioning.

**Protective Measures by Phase:**

| PHASE | PROTECTIVE MEASURES |
|---|---|
| **Shipping and Receiving** | • Use sealed packaging and vent covers<br>• Client and GC to trend toward transport on plastic pallets wherever feasible to minimize splinter-based FOD<br>• Inspect and photo-document gear condition at arrival |
| **Staging and Storage** | • Raise equipment on blocks or platforms to avoid ground contact<br>• Keep gear in sealed, labeled locations away from demo or fabrication areas<br>• Enclose with polywrap or zippered tent if long-term storage is needed |
| **Work in Progress** | • Covers must be replaced after every shift or work pause<br>• If temporarily left open, enclosures must be fully sealed with plastic or magnetic covers |

## EQUIPMENT PROTECTION – EXTERIOR

Exterior or prefabricated equipment requires its own layer of protection due to exposure to weather, logistics handling, and remote staging risks.

### Standards for Exterior Gear:

- **Visual Inspection on Arrival:**
  - Check exterior for dents, corrosion, or punctures
  - Open enclosures to confirm no packaging material or debris remains inside
- **Packaging**:
  - Use UV-rated, waterproof plastic wrap
  - Seal all openings with tamper-evident tape or mechanical closure
- **Documentation:**
  - Take photos upon arrival and upload to server with equipment ID and location
  - Include delivery checklist and transport chain of custody form
- **Staging Requirements:**
  - Shade structures, fenced containers, or pop-up shelters for sensitive gear
  - Install signage for identification, restricted access, and hazard warnings
  - Assign staging zones based on equipment type and proximity to construction activities

## HOUSEKEEPING

A clean site is a safe site. Housekeeping is an all-hands responsibility. Clean sites reduce injuries and eliminate arc flash contributors.

### Responsibility and Leadership

Each trade is required to designate a FOD Captain per shift. This person leads cleanliness efforts, ensuring that work areas remain clean and organized and that tools and materials are stored appropriately. FOD Captains are responsible for verifying that gear is not used for tool storage and that work zones are free from debris at all times.

### Daily Expectations

Crews must sweep or vacuum their work zones before and after every shift. Tools should be returned to labeled carts, packaging must be removed immediately after installation, and all panel covers must be replaced after access. This practice keeps critical areas free from hazards and ensures compliance with FOD control.

### Weekly Practices

Formal walkdowns are conducted with the GC and trade leads to inspect work zones, reinforce positive behaviors, and address non-compliance. These sessions double as field-level training opportunities.

### Containment and Prevention

Controlled entry points must be equipped with walk-off mats, tac-mat or Skudo zones , and boot scrubbers. HEPA vacuums are to be used instead of brooms, particularly near live or sensitive equipment. Whenever cutting, sanding, or overhead work occurs, containment barriers should be installed to trap dust and prevent FOD migration.

### Organization of Tools and Materials

Tools and materials must be clearly labeled, with carts and bins assigned by trade. Shared or mixed-use storage should be avoided to reduce confusion. Material storage zones must be documented and laid out to support safe access and reduce trip hazards.

### Non-Negotiables

Eating, drinking, or leaving personal items in or around electrical gear is prohibited. No materials or tools may lean against equipment. Cutting or grinding within electrical rooms is strictly forbidden.

# FOD (Foreign Object Debris) Marshal roles typically include:

Inspecting and ensuring that work areas, especially construction or manufacturing sites, are free from foreign objects that could cause damage or safety hazards. The FOD Marshal is typically a GC employee and is part of the electrical team on the project (Not necessarily is this an added member to the team but rather a qualification and expectation of the GC's field electrical leadership, i.e. GC Electrical Superintendent).

- Coordinating with project teams to implement FOD control measures.
- Conducting regular audits and inspections.
- Training personnel on FOD awareness and prevention.
- Reporting and documenting any FOD incidents.

A FOD (Foreign Object Debris) Marshall must work very closely with site safety teams, subcontractors and project management to ensure timely escalation of hazards and clear reporting.

- Emphasize proactive risk elimination
- Data-driven follow-up and strong subcontractor engagement:
- Daily coordination with site safety team
    - Attend morning briefings to review any new FOD observations and agree on immediate corrective actions.
    - Use the Hierarchy of Control mindset to propose elimination or engineering controls first, then administrative and PPE measures as needed.
- Structured reporting and escalation to project management
    - Log every FOD find in the site safety database or reporting tool—capturing location, root cause, corrective action and responsible party.
    - Trigger an "immediate escalation" when a high-potential debris incident is identified (i.e., any debris that could cause serious injury or equipment damage) and notify the Project Sponsor and Safety Director by email or SMS within one hour.
- Engaging subcontractors in FOD prevention
    - Hold weekly toolbox talks with all subcontractor forepersons to review recent FOD trends and reinforce best practices.
    - Require subcontractors to submit their own FOD control plans and evidence of daily housekeeping checks; integrate these into your weekly performance follow-up metrics.
- Performance monitoring and continuous improvement
    - Include FOD metrics (number of finds, time to resolution, repeat locations) in the site's monthly safety performance report, reviewed by Business Unit leadership and the Group Leadership Team.
    - Feed lessons learned back into the Health & Safety Road Map for both the local site and global template updates.

By following this structured approach, grounded in proactive elimination of hazards, clear lines of communication, and robust data analysis you'll ensure foreign-object debris is managed efficiently and safely across your project.

# FOD (Foreign Object Debris) Captain on a project:

This role is typically a Trade Partner site leader or an OFCI field leader that is part of the site electrical team, alternately this responsibility can be given to all electrical foreman with all the described inspection and reporting listed below:

### Leadership & Coordination

- Act as the single point of contact for all FOD control activities on site.
- Convene and lead daily FOD briefings with safety, site engineering, logistics and subcontractor teams.

### Inspection & Monitoring

- Conduct scheduled and random site walks to identify, tag and remove debris or foreign objects.
- Use digital checklists or mobile apps to record findings, assign corrective actions and track closure.

### Escalation & Reporting

- Classify each FOD finding by risk level; immediately escalate high-risk items (those posing potential equipment damage or personnel injury) to the Safety Director and Project Manager.
- Produce and distribute a weekly FOD dashboard showing trends, time-to-resolution and repeat locations.

### Subcontractor Engagement

- Integrate FOD requirements into subcontractor pre-mobilization meetings and ensure each trade has a site-specific FOD plan.
- Hold weekly toolbox talks with subcontractor crews to review recent FOD incidents, trends and prevention techniques.

### Training & Culture Building

- Deliver FOD-awareness training to all new site arrivals, including office staff, tradespeople and visitors.
- Promote “see it, own it, solve it” culture—empowering every team member to pick up or report debris on sight.

### Continuous Improvement

- Analyze FOD incident data to identify “hot-spot” locations or processes that drive debris generation.
- Recommend and pilot engineering or administrative controls (e.g., mesh flooring, dedicated debris containers, housekeeping rotations) to eliminate root causes.

### Integration with Safety & Quality Systems

- Align FOD controls with the broader site Safety Management System and quality checkpoints.
- Feed FOD metrics into the monthly Safety & Quality performance review with the Project Leadership.

FOD Captain ensures consistent debris control, strengthens cross-functional collaboration, and builds a proactive safety culture throughout the project.

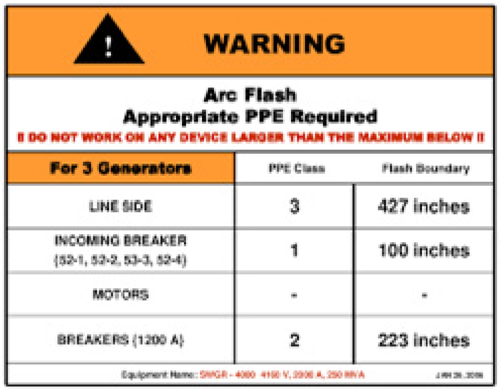
# Energy Isolation / LOTO
## *(Lock Out, Tag Out)*

## Arc Flash Safety

***Arc flashes are sudden electrical explosions caused by a fault through the air, releasing intense heat, light, sound, and pressure. These incidents can cause severe injuries such as third-degree burns, blindness, hearing loss, or even death. The extreme temperature from an arc flash can reach up to 35,000°F, igniting clothing and causing fires. Preventing arc flash incidents is a top safety priority in mission critical projects and requires meticulous planning, engineering controls, personal protective equipment (PPE) , qualified personnel, and detailed work procedures.***

**ARC FLASH HAZARD ANALYSIS**
A comprehensive Arc Flash Hazard Analysis must be performed for all energized electrical systems as part of the High-Risk Activity (HRA) Management Plan. The goal is to identify potential arc flash hazards and implement effective mitigation measures. The analysis should include:

- **Incident Energy Calculations:** Assess the amount of energy (in cal/cm$^2$) that could be released during an arc event at different points in the electrical system. This determines the necessary PPE and safe approach distances.
- **Arc Flash Boundary Identification:** Prior to First Time Energization (FTE), Establish the arc flash boundary, which is the minimum distance from an energized component where a person could receive second-degree burns from an arc flash .
- **Labeling Requirements:** All electrical equipment must be labeled in accordance with NFPA 70E standards. Labels should display the arc flash boundary, incident energy, PPE requirements, and the equipment's voltage.
- **Documentation and Review:** Results of the analysis must be documented, reviewed periodically, and updated after any modifications to the electrical distribution system.



WARNING

Arc Flash
Appropriate PPE Required
!! DO NOT WORK ON ANY DEVICE LARGER THAN THE MAXIMUM BELOW !!

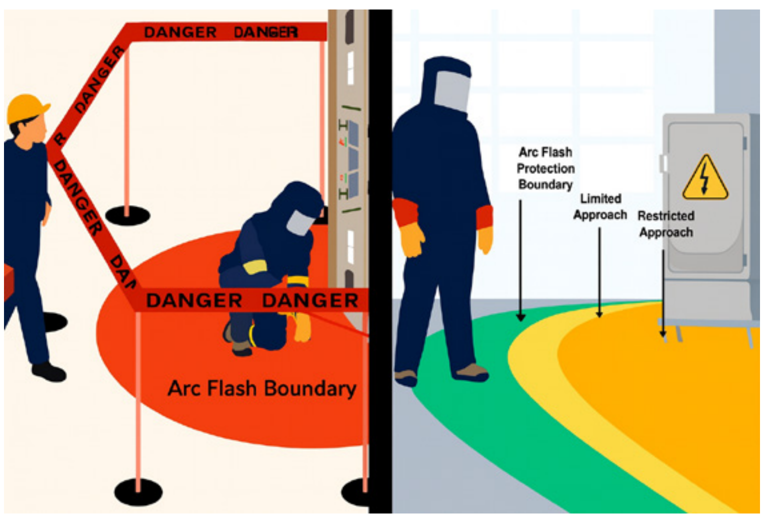
| For 3 Generators | PPE Class | Flash Boundary |
|---|---|---|
| LINE SIDE | 3 | 427 inches |
| INCOMING BREAKER (52-1, 52-2, 53-3, 52-4) | 1 | 100 inches |
| MOTORS | - | - |
| BREAKERS (1200 A) | 2 | 223 inches |

*Sample arc flash label*

## ARC FLASH BOUNDARIES

There are three critical boundaries defined by NFPA 70E to ensure personnel safety when working around energized equipment:

- **Limited Approach Boundary:** The distance from exposed energized conductors within which only qualified personnel or escorted unqualified persons may enter. It marks the point where increased risk begins. This limited approach shall be established by physical barrier to protect general personnel from boundary violation.
- **Restricted Approach Boundary:** A closer boundary where only qualified workers using proper PPE and work techniques may enter. Special training and risk assessments are required to work within this zone. Qualified personnel will ascertain this boundary from equipment labels.
- **Arc Flash Boundary:** This boundary is calculated based on incident energy analysis. Any person crossing this line must wear PPE appropriate for the level of arc flash energy present. Qualified personnel will ascertain this boundary from equipment labels.

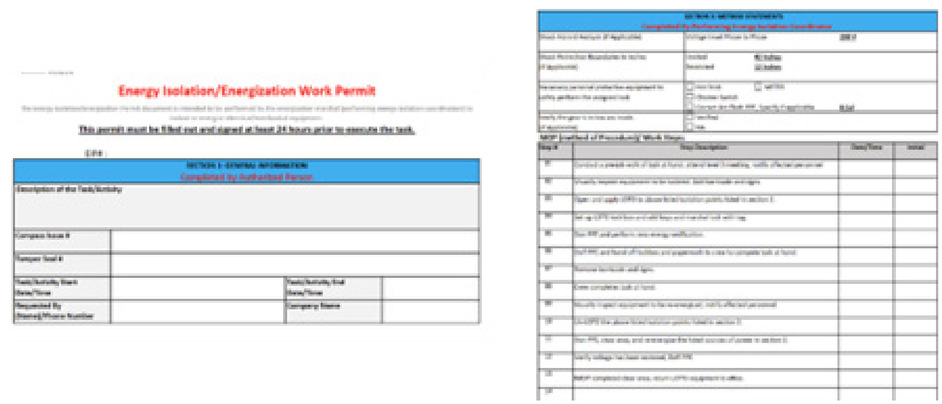



*Examples of Arc Flash Boundaries*

## ENERGY ISOLATION PERMIT (EIP)

An Energy Isolation Permit (EIP) is a mandatory safety document required whenever personnel work near or within restricted or arc flash boundaries. The EIP ensures that all energized work is justified and carried out safely with proper controls in place. The permit must be reviewed and approved by the Authorizing Authority Energy Marshal and include:

- **Justification for Energized Work:** A clear explanation of why the equipment cannot be de-energized.
- **Hazard Risk Assessment:** Description of potential hazards, energy sources, and control methods.
- **PPE Requirements:** Confirmation that appropriate PPE has been selected based on the hazard analysis.
- **Rescue Plan:** Emergency response procedures, including rescue roles and required equipment.
- **Authorization:** Signatures from the Performing Authority, Electrical Coordinator, and Authorizing Authority, confirming approval to proceed.




*Example of Energy Isolation/Energization permit (Sections 1 & 2)*

**SECTION 2: LOCK-OUT TAG-OUT INFORMATION**

Completed By Performing Energy Isolation Coordinator

| Equipment To be Isolated | Applied LOTO Device # (Breaker, Switch, etc.) | LOTO Apply Time | Lock Number | Initial | LOTO Remove Time | Initial | Device Status (Open/Closed) |
|---|---|---|---|---|---|---|---|
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*Example of Energy Isolation/Energization permit (Section 3 & 4)*

**SECTION 4: LOCK-OUT TAG-OUT INSTALL CHECK LIST**

Completed By Performing Energy Isolation Coordinator

| STEPS | INITIALS | STEPS | INITIALS |
|---|---|---|---|
| 1) Ensure all isolating devices will accept LOTO | | 2) Review procedures | |
| 3) Notify affected workers | | 4) Isolate energy source(s) (document time above) | |
| 5) Install LOTO(s) (document lock # above) | | 6) Verify equipment is de-energized | |

**HOLD POINTS**

A. Stop work if any unexpected energy is found.
B. Stop work if equipment has been modified since the authorization was issued.
C. Stop work if any unanticipated circuits or equipment are found to be affected by the LOTO.

Note: Work is not authorized beyond any hold point until the issue has been resolved.

**NOTES**

* ALL ELECTRICAL CONDUCTORS SHALL BE REGARDED AS ENERGIZED UNTIL PROVEN SAFE. *

*LIVE WORK METHODS AND PPE SHALL BE USED WHILE VERIFYING THE ABSENCE OF ENERGY. *

**Permit Approval**

| | | | |
|---|---|---|---|
| Performing Energy Isolation Coordinators | Name: | Signature: | Date: |
| GC Authorizing Energy Marshal | Name: | Signature: | Date: |
| GC Field Supervisor | Name: | Signature: | Date: |

Note: GC Safety or energy marshal signature to be obtained during L3 meeting.

## PERSONAL PROTECTIVE EQUIPMENT (PPE) FOR ARC FLASH

The correct use of PPE is critical to protecting workers from arc flash incidents. PPE must be selected based on the incident energy analysis and include:

- **Arc-Rated Clothing:** Flame-resistant (FR) garments rated to withstand the calculated incident energy.
- **Arc-Rated Hood or Face Shield with Balaclava:** Provides head, face, and neck protection from thermal effects.
- **Insulated Gloves and Sleeves:** Protect hands and arms from electric shock and burns.
- **Protective Footwear:** Electrically rated boots with non-conductive soles.
- **Eye and Hearing Protection:** Safety glasses or goggles and hearing protection must be worn under the hood.

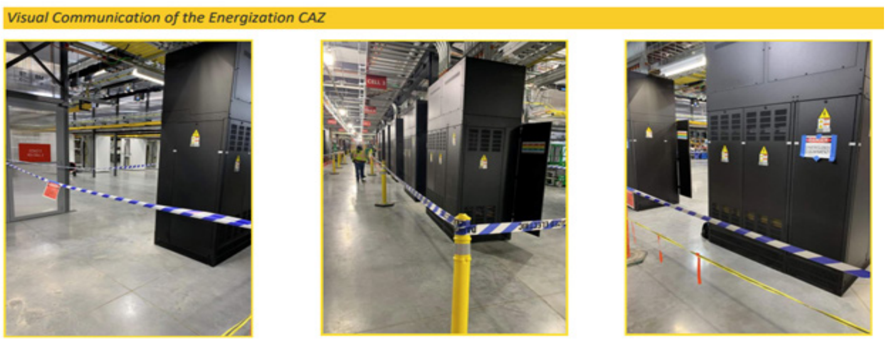
PPE must be inspected before each use for damage or contamination. Defective equipment must be replaced immediately.

## SAFE WORK PRACTICES

The following safe work practices help minimize the risk of arc flash:

- **De-energize Whenever Possible:** Work should only be performed on live equipment when absolutely necessary.
- **Establish Electrically Safe Work Conditions:** Follow LOTO procedures to ensure that all energy sources are isolated and verified to be at zero voltage.
- **Use Barriers and Signage:** Maintain physical barriers and clear warning signs around energized areas.
- **Verify Absence of Voltage:** Use approved test instruments to verify that no voltage is present before work begins.
- **Buddy System:** Workers must not work alone when performing energized tasks.

These practices must be followed at all times and documented as part of the work plan.




*Example Visual Communication of the Energization*

## ARC FLASH MITIGATION STRATEGIES

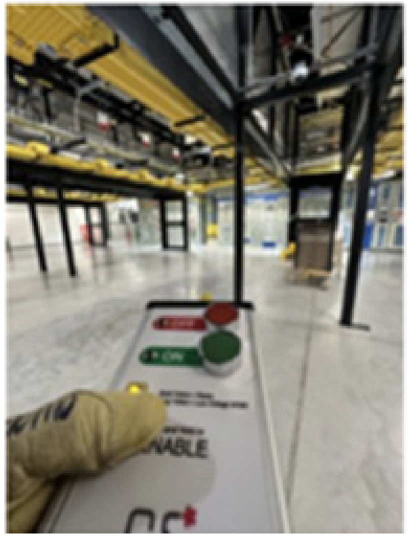
Strategies to reduce the likelihood and severity of arc flash incidents include:

### Pre-Energization Planning and Setup

- Create a detailed Electrical Method of Procedure (EMOP) that breaks down all steps of planned energization.
    - Use a sign off column to initial each step as completed to confirm planned sequencing is followed
- Review EMOP in a LEVEL 2 pre planning meeting to align with all parties
- Create and distribute a logistics map that highlights all affected areas on site. Include CAZ locations, nurses trailer locations, safety / first aid stations.
- Walk the path of energization, verify all equipment is closed up and tamper sealed, and clear all non-essential personnel from the area.
- Final LEVEL 3 meeting with the energy marshal team to confirm no changes are needed and you are ready to proceed with the EMOP activities.

## Remote Switching

- Use “Chicken Switches ” or similar remote racking tools to operate live electrical equipment from a safe distance.

*Breaker remote switching, using a manufacturer-provided or customized CBS Arcsafe solution, ensures safe and efficient operation of MV circuit breakers. It is especially useful for critical equipment like UMS, MSB, and UDS during initial energization – a period prone to arcing incident or equipment failures.*

## Preventative Maintenance

- Regular maintenance reduces the chance of equipment failure. Document inspections and keep all devices clean and sealed.

## Upgraded Protective Devices

- Replace old breakers with current-limiting devices to reduce available incident energy.

## Equipment Delivery and Preservation

- Whether delivered to an offsite storage facility or directly to the jobsite, all electrical equipment must be received and preserved with care. This includes documented inspections upon delivery, elevated storage off the ground, and detailed preservation plans with vendor buy-in that address climate control and pest protection. Environmental monitoring tools (e.g., Govee, Pillar) should be used to track temperature, humidity, and dust levels.

## Foreign Object and Debris (FOD) Control

- To prevent injuries and equipment damage during energization, strict FOD controls must be enforced. These include keeping access covers in place, performing cutting and drilling outside of gear whenever possible, maintaining clean work areas, using plastic and rubber covers, prefabricating materials, and isolating work zones with barricades. Environmental cleaning tools like vacuums and air scrubbers should be used as needed.

## FOREIGN OBJECT AND DEBRIS (FOD) CONTROL

FOD is anything left behind that is not a manufactured or construction installed component of the equipment. Below is the site-specific program to control the hazards associated with foreign objects and debris left behind during the installation and testing of equipment that could result in injury or property damage during energization of equipment.

- Access covers to equipment to remain on all gear when possible and work is not being completed.
- Drilling and cutting shall be conducted outside of equipment at all-time unless deemed unfeasible.
- Cutting operations will be performed at a designated cutting station.
- Always utilize the proper tools for the job.
- All drilling in equipment shall have FOD countermeasures in place prior to activity commencing.
- Clean work areas shall be always maintained with daily clean ups.
- No storage of tools or excess materials in equipment.
- Use plastic covers for the equipment's to keep the equipment clean.
- Use plastic covers for all parts of equipment that may remain open during construction process prior to install.
- Keep the equipment doors closed unless work is being performed.
- Use rubber blankets to protect equipment from tools and debris.
- Vacuum clean equipment's as needed.
- Air scrubbers will be utilized as needed.
- Prepackaged lugs/parts for each gear.
- Prefabricated materials as possible.
- Barricades shall be used to prevent personnel not directly involved with the installation of the gear from accessing the equipment

### Temp Power Energization and Isolation Process

- Develop detailed QC checklist and document with photo evidence that all temporary power equipment is installed correctly and meets all national codes and standards
- LEVEL 2 pre planning meeting to review all energization and isolation activities
- Create detailed electrical method of procedure (EMOP)
- Include all temporary power scopes in site lock out plan. Utilize digital lockout/tagout system (eg AutoLOTO) on all projects where applicable to manage both permanent and temporary power facilities. Regular one-line and panel schedule updates are required from EOR.
- Include hard barricades around all construction power units. Consider skid and mobile units to aid in flexibility for construction.

## ARC FLASH TRAINING AND COMPETENCY

Training is essential to ensure all personnel working near energized systems understand the risks and know how to protect themselves and others. Training programs must cover:

- **Hazard Recognition:** How to identify signs of arc flash risks and associated hazards in the field.
- **Use and Care of PPE:** Instruction on how to correctly don, use, inspect, and maintain arc flash-rated PPE.
- **Work Procedures:** Safe work practices including LOTO, boundaries, and verification techniques.
- **Emergency Response:** Actions to take in the event of an arc flash, including how to initiate rescue and administer first aid

and how to use a fire extinguisher. (Note that only Halotron Fire Extinguishers should be utilized around electrical gear).

Training must be documented, repeated at regular intervals, and updated when new equipment or procedures are introduced.

## EMERGENCY RESPONSE AND RESCUE PLAN

Every job site must have a detailed, *Site-Specific Emergency Response and Rescue Plan* to deal with arc flash and electrical emergencies. The plan should include:

- **Rescue Methods:** Guidelines for safe extraction of injured personnel without placing rescuers at risk.
- **Rescue Equipment:** Availability of appropriate tools such as insulated rescue hooks, fire blankets, and AEDs.
- **Personnel Designation:** Identify trained rescuers on each shift with certifications in first aid, CPR, and electrical rescue.
- **Drills and Simulations:** Regular practice sessions to test the team's readiness and improve response time. Timeline for consideration would be ahead of Initial building FTE and annually as required by NFPA 70E.

## AUDITS AND PROGRAM REVIEW

Plans must be accessible, reviewed regularly, and updated as site conditions change. Routine audits of the Electrical Safety Program and Energized Electrical Work Permits are essential for verifying compliance, uncovering gaps, and promoting continuous improvement. Lessons learned and best practices should be captured and integrated into future procedures to enhance safety and performance across all projects.

## ELECTRICAL SAFE WORK PROCEDURES

Certain electrical tasks require formal documentation, planning, and approvals to ensure safety. These tasks include:

- **Working on Systems Over 50V:** Any maintenance or service involving systems greater than 50 volts AC or DC requires written procedures and appropriate PPE.
- **Entering Electrical Boundaries:** Crossing into limited, restricted, or arc flash boundaries requires a risk assessment, proper training, PPE, and possibly an Energy Isolation Permit.
- **Ground Disturbance or Wall Penetration:** When working near underground or concealed electrical installations, verification and marking of locations must be completed before starting work.
- **Battery Charging and Storage:** Procedures must address the handling of high-voltage batteries and the associated chemical and electrical hazards.
- **Temporary Power Installation:** Planning must include energization, de-energization, isolation plans, digital lockout/tagout system (eg. AutoLOTO) usage, and inspection logs.
- **Resetting Breakers or Removing Panels:** All such actions must be performed by qualified individuals with proper hazard identification and PPE.

# Energy Marshal Program

*The Energy Marshal Program is a core component of mission-critical electrical safety, designed to ensure strict adherence to energy isolation protocols across all phases of construction, commissioning, and maintenance. Energy Marshals are highly trained professionals tasked with overseeing the planning, execution, verification, and auditing of electrical isolation tasks, particularly in high-risk environments such as data centers.*

### PURPOSE AND IMPORTANCE OF ENERGY ISOLATION MANAGEMENT

The primary role of the Energy Marshal is to prevent the accidental release of hazardous energy that could lead to injury or equipment damage. This energy could be in the form of electrical, mechanical, hydraulic, pneumatic, chemical, thermal, or other types of hazardous energy. This is accomplished by enforcing safe work practices, verifying isolation points, ensuring compliance with all lockout/tagout (LOTO) procedures, and coordinating among stakeholders.

### ENERGY MARSHAL KEY RESPONSIBILITIES

- **Program Oversight:** Manage the implementation of the energy isolation strategy across the project. Ensure policies, procedures, and tools are properly integrated and followed.
- **Permit Review:** Review and approve Energy Isolation Permits (EIPs) to ensure proper documentation, risk assessments, and emergency procedures are in place.
- **Field Verification:** Personally verify that isolation points are properly locked out and tagged, and that the equipment is de-energized and tested before work begins.
- **Coordination with Teams:** Act as a liaison between project managers, electricians, safety officers, and vendors to confirm alignment on energy isolation protocols.
- **Training and Communication:** Conduct training sessions on LOTO procedures, arc flash hazards, and isolation planning. Simplify complex technical content for broader understanding.

### COLLABORATION AND LEADERSHIP

Energy Marshals play a proactive role in fostering a safety-first culture. They participate in pre-task planning meetings, HRA sessions, and MOP reviews. They ensure subcontractors are qualified and informed, and that no work begins until all isolation requirements are met. The Marshal’s presence during high-risk energization or de-energization work is essential to verify readiness and enforce discipline.

### ENFORCEMENT AUTHORITY

Energy Marshals are empowered with Stop Work Authority. If any aspect of the energy isolation plan is violated, or if unsafe conditions are observed, they are responsible for halting the work until the issue is resolved. Their oversight ensures that no shortcuts are taken and that all personnel are protected.

### CONTINUOUS IMPROVEMENT

Marshals conduct regular audits, track lessons learned and provide feedback to improve procedures. Their insights are crucial for updating safety documentation and refining project-wide isolation strategies.

*The Energy Marshal is not just an enforcer of safety but a strategic partner in project success, combining technical knowledge, leadership, and communication skills to ensure that energy isolation procedures are safe, compliant, and effective.*

### ROLE OF ENERGY MARSHAL IN VENDOR COORDINATION

Energy Marshals play a critical role in ensuring that all subcontractors and vendors align with site-specific energy isolation protocols. Their leadership ensures that third-party work does not introduce risk to personnel or the system.

Key Responsibilities:

- **Pre-Mobilization Review:** Vendors must submit detailed work plans that clearly outline energy isolation steps. Energy Marshals assess these for completeness and compliance.
- **Training Verification:** Energy Marshals confirm that vendor personnel are properly trained, certified, and understand site-specific requirements, including PPE, LOTO, and emergency procedures.
- **Coordination and Communication:** Vendors are integrated into the overall site energization strategy through planning meetings, shared documentation, and isolation maps.
- **Compliance Audits:** Regular spot checks and audits are conducted to confirm that vendor teams are executing the plan safely and correctly. Non-compliance is documented and addressed immediately.
- **Stop Work Enforcement:** Marshals have the authority to stop vendor work at any time if isolation procedures are violated or if unsafe behavior is observed.

By maintaining close oversight, Energy Marshals ensure that vendors do not operate in isolation but are fully integrated into the site's comprehensive energy safety framework.

### HAZARD RISK ASSESSMENT (HRA) PLANNING MEETINGS

Hazard Risk Assessment (HRA) planning meetings are structured in three escalating levels of detail and formality to align teams before beginning high-risk electrical work.

- **Level 1 (L1) – Initial Risk Identification -** This session introduces the planned work, its associated hazards, and preliminary mitigation strategies. Participants include safety leads, supervisors, and relevant stakeholders. It sets the foundation for deeper reviews.
- **Level 2 (L2) – Detailed Planning -** This meeting involves a thorough review of the Method of Procedure (MOP), lockout points, system redundancies, and team assignments. Visual aids such as one-line diagrams and isolation maps are reviewed, and logistics such as signage, barricades, and CAZ (Controlled Access Zones) are confirmed.
- **Level 3 (L3) – Final Execution Readiness -** Conducted immediately before the start of work, this meeting ensures that all documentation is current, equipment is verified in the field, and teams are briefed and ready. Final approvals from the Energy Marshal and Authorizing Authority are obtained here.

All HRA meetings must be documented with meeting minutes and sign-in sheets. Assignments must be clearly communicated and confirmed by participants. Meetings should be documented, and responsibilities clearly assigned to all participants.

### HRA AUDIT

HRA audits ensure that energy isolation activities follow established protocols and that risk mitigation plans are executed as intended. Audit Responsibilities:

- **Daily:** The Safety Manager performs walk-through inspections to verify that safety controls are in place, documentation is accessible, and any changes in scope are accounted for.
- **Weekly:** The Energy Marshal performs targeted audits of electrical tasks and isolation procedures, focusing on the consistency of lockout applications and alignment with the approved MOP.
- **Quarterly:** A comprehensive review of the HRA program is conducted to evaluate program effectiveness, update procedures, and document lessons learned.

**HRA AUDIT - RESPONSIBILITY MATRIX**

| ROLE | TASK | FREQUENCY |
|---|---|---|
| Safety Manager | One Electrical HRA Audit per day | Daily |
| Energy Marshal | One Electrical HRA Audit per week | Weekly |
| Energy Marshal | One Electrical Isolation Audit per week | Weekly |
| Energy Marshal | Quarterly Energy Marshal plan review | Quarterly |

Audits must include validation of worker qualifications, review of active permits, interviews with personnel, and field verification of isolation points. Findings are documented and shared during safety meetings to drive continuous improvement.

### CONTRIBUTION TO THE LOTO PROCESS

The Lockout/Tagout (LOTO) process is the primary means of energy isolation, and the Energy Marshal plays a key role in overseeing and ensuring the correct implementation of this process. The Energy Marshal's responsibilities in the LOTO process include:

- **Issuing Permits:** The Energy Marshal ensures that energy isolation permits are issued for all electrical and mechanical work, reviewing and approving each permit before work begins.
- **Ensuring Proper Application of LOTO:** Verifying that all energy-isolating devices are properly applied, including locks and tags, and that workers are following the correct procedures for isolating energy sources.
- **Monitoring Compliance:** The Energy Marshal actively monitors the worksite to ensure that workers follow LOTO protocols and that no unauthorized personnel are allowed to access energy-isolated equipment.
- **Real-Time Tracking:** In some cases, the Energy Marshal utilizes digital lockout/tagout tools like AutoLOTO to digitally track the status of energy isolation devices, ensuring that locks are applied correctly and that no energy is inadvertently reintroduced to the system.

By ensuring the proper execution of LOTO, Energy Marshals help mitigate the risks associated with energized systems and maintain a safe working environment.

### STOP WORK AUTHORITY

All personnel on-site have the duty and authority to initiate a Stop Work if they observe unsafe conditions or a deviation from the approved procedures. This culture of shared accountability is central to maintaining site safety.

**Energy Marshal's Role:**

- **Immediate Intervention:** The Energy Marshal will promptly halt operations if any deviation from energy isolation protocols is observed.
- **Root Cause Analysis:** A rapid evaluation is conducted to identify the reason for the deviation, involving the affected team, supervisors, and safety leads.
- **Corrective Action:** Isolation plans, training, or equipment are updated before resuming work. In some cases, retraining or re-approval of permits may be required.
- **Documentation and Communication:** All stop work events must be logged and communicated during safety stand-downs or toolbox talks as part of a learning culture.

The Stop Work Authority ensures that all personnel - from entry-level workers to leadership - feel empowered and responsible for site safety.

### ENERGY ISOLATION AUDITS

Energy Marshals conduct routine and event-driven audits of isolation activities to validate execution against the approved plan. Audits are both proactive (to catch issues before they happen) and retrospective (to identify process improvements).

**Audit Components:**

- **Documentation Review:** Check all active EIPs and MOPs for completeness and consistency with the work being performed.

- **Physical Verification:** Confirm that locks, tags, and devices are applied correctly and match the approved isolation plan.
- **Condition Assessment:** Ensure that equipment is properly preserved, closed, and labeled, with no evidence of tampering or unauthorized access.
- **Personnel Check:** Interview workers to confirm they understand the scope, boundaries, and steps of the task.

Audit results feed into site-wide metrics and safety dashboards, contributing to trend analysis, training improvements, and procedural refinement.

## COMMUNICATION OF TECHNICAL CONCEPTS

Energy Marshals are tasked with communicating complex technical concepts related to energy isolation and electrical safety to a variety of audiences, from skilled electricians to project managers and subcontractors. Clear communication is essential for ensuring that all personnel understand the safety protocols and the importance of energy isolation.

### Energy Marshals communicate by:

- Simplifying Complex Procedures: Breaking down technical procedures into clear, actionable steps that all personnel can follow.
- Providing Training: Leading safety training sessions that help workers understand the risks of electrical work, how to implement energy isolation, and the importance of proper lockout/tagout procedures.
- Ensuring Clarity in Documentation: Writing clear and concise MOPs and energy isolation procedures that everyone on the site can easily reference.

By translating technical safety protocols into clear communication, Energy Marshals ensure that everyone is on the same page and understands the importance of energy isolation.

## METHOD OF PROCEDURES (MOPS)

Method of Procedures (MOPs) are structured documents that define exactly how electrical systems will be de-energized and re-energized. They ensure that complex isolation activities are carried out consistently, safely, and in coordination with all parties.

MOPs serve as the blueprint for safely isolating energy sources and are essential for:

- **Standardizing Work:** MOPs ensure that each energy isolation task is performed consistently and safely, following the same steps every time.
- **Defining Safety Measures:** They specify the required PPE, tools, and energy isolation steps for each task, ensuring that no steps are missed.
- **Reducing Risk:** By following MOPs, workers can minimize the risk of accidents and ensure that energy isolation is performed correctly.

Energy Marshals play a vital role in ensuring that MOPs are created, reviewed, and adhered to, ensuring safe work practices across all projects.

### Key Elements of an MOP:

- **Scope and Objectives:** Clearly define what equipment is affected, what work will be done, and why.
- **Step-by-Step Instructions:** Detail each task in logical sequence with sign-off checkpoints, include any interdependence between MOP’s for visibility and clarity of sequence.
- **Verification Requirements:** Include testing requirements, PPE checks, tool inspections, and labeling confirmation and tamper seal compliance (see Tamper Seals).
- **Assigned Roles:** Define responsibilities, including the Energy Marshal, Performing Authority, Safety Manager, and Qualified Electrical Workers.

- **Contingency and Exit Plans:** Include steps for safe back-out if conditions change or problems arise.

MOPs are reviewed during HRA meetings and must be updated whenever equipment, personnel, or work scopes change

## Energization Method Of Procedure

*The Energization Method of Procedure (EMOP) document is intended to be used for the initial energization of electrical distribution equipment.*

| | | | |
|---|---|---|---|
| **Go Date:** | 18 Nov 2020 | **Consent Signature:** | Matthew Vetter |
| **Summary:** | Initial energization / start-up of XXXX-4A-FAP01. | | |
| **Safety Requirements:** | ☐ Standard OSHA, Microsoft, Turner, and CST safe practices.<br>☐ Appropriate PPE per arc flash label. | | |
| **Risks:** | 1. Initial energization increases chance of electrical failure.<br>2. Arc Flash. | | |

**Detailed Procedure:**

| *Initial, date, and time boxes must be completed (N/A the boxes of steps that do not apply)* | | **Initial** |
|---|---|---|
| 1 **Conduct a thorough pre job brief** covering all details of the procedure. | | |
| 2 **Install restricted access zone** around equipment to be energeized. | | |
| 3 **Verify** all access panels and doors are latched/fastened closed. | | |
| 4 **Verify** xxxx-4A-FAP01 Yellow Tag is signed and dated. | | |
| 5 **Remove** Controlling LOTO of xxxx-4A UPS01 PNL01 CB05 breaker. | | |
| 6 **Close** DSM09-4A UPS01 PNL01 CB05 breaker. | | |
| 7 **Verify and Document.** | | |
| **Equipment** | **Voltage** | |
| xxxx-4A UPS01 PNL01 | | |
| 8 **Verify** all access panels and doors are latched/fastened closed. | | |
| 9 **Notify** equipment vendor to complete equipment startup. | | |
| 10 EMOP completed. | | |

*Sample MOP step breakdown or checklist*

**INTERPRETING LINE DRAWINGS AND SYSTEM REDUNDANCIES**

Accurate interpretation of line drawings and understanding system redundancies are essential for identifying energy sources and developing effective energy isolation plans. The Energy Marshal must:

- **Interpret Line Drawings:** These drawings help identify energy isolation points, showing how energy flows through systems and highlighting potential hazards.
- **Understand System Redundancies:** In systems with redundancies (e.g., backup generators or power feeds), Energy Marshals ensure that all potential energy sources are isolated and that backup systems are considered when isolating energy.

By understanding and interpreting these diagrams, Energy Marshals can design effective lockout/tagout systems that ensure all potential energy hazards are addressed.

# FOK (First of Kind) Inspections

*First of Kind (FOK) inspections are a critical quality assurance milestone. They are intended to verify that the first instance of a specific equipment type has been installed correctly, tested thoroughly, and meets project expectations. Conducting FOK inspections ensures that installation deficiencies are corrected before full production begins and reduces delays, rework, and costly surprises during commissioning.*

## Objectives:

- Prevent systemic errors across multiple installations.
- Confirm compliance with contract drawings and submittals.
- Ensure that all installation details (hardware, clearances, torque specs, terminations) meet industry standards.
- Align all project stakeholders on acceptable quality levels.

## Stakeholder Participation:

- **Electrical Contractor (EC):** QC Manager and Foreman
- **General Contractor:** QA Representative and Electrical Subject Matter Expert (SME)
- **Equipment Vendors:** To validate their equipment's install and commissioning conditions
- **Commissioning Agent:** To align early on verification and turnover expectations
- **Owner's Representatives and A/E Teams:** Optional, but valuable for firsthand input

## Inspection Components:

- Review of documentation (BOL, torque logs, checklists, test reports)
- Visual inspection of wiring, labeling, lashing, and separation
- Grounding verification (type and torque of bolts, lug installation)
- Status of environmental controls, labels, nameplates, and tags
- Confirm mockup-specific lessons are documented and communicated

A FOK Inspection Form is used to capture results, photos, best practices, and signatures from all parties. This record sets a precedent for subsequent installations. See Appendices for sample forms.

***See Exhibit F FOK Inspection Checklist***

First of Kind (FOK)
**Field Inspection Checklist**

| OBSERVATION DETAILS | | | | | |
|---|---|---|---|---|---|
| Project Name and #: | Location/Area: | Observation Type: | Observation Date: | Reported By: | Reviewed With: |
| NAME | COLO# / Cell# | FOK installation | Date | Name / Company | GC: Name / Company<br>EC: Name / Company<br>Vendor: Name / Company<br>EOR: Name / Company<br>MSFT: Name |

| Equipment Name | GC | EC | Vendor | EOR | MSFT | Date | Comments/Observations: |
|---|---|---|---|---|---|---|---|
| Terminations per specifications and equipment submittals | | | | | | | |
| Conductor supports per equipment shop drawings and reviewed by vendor | | | | | | | |
| Attachments to equipment (chimneys, side cars, top hats) reviewed by vendor, EOR | | | | | | | |
| EPMS / BAS biscuit location | | | | | | | |
| No signs of water infiltration ** if signs of water infiltration the source must be identified | | | | | | | |
| Labels installed – correct type and locations | | | | | | | |
| No signs of FOD / metal shavings ** if metal shavings are found the source must be identified | | | | | | | |
| Verify no physical damage | | | | | | | |
| Confirm proper grounding | | | | | | | |
| Confirm proper anchoring | | | | | | | |
| Note: Add equipment specific items based on submittals, TSBs issued, etc. that need to be reviewed during FOK inspection. | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |

Page 1

*Sample FOK Inspection Form*

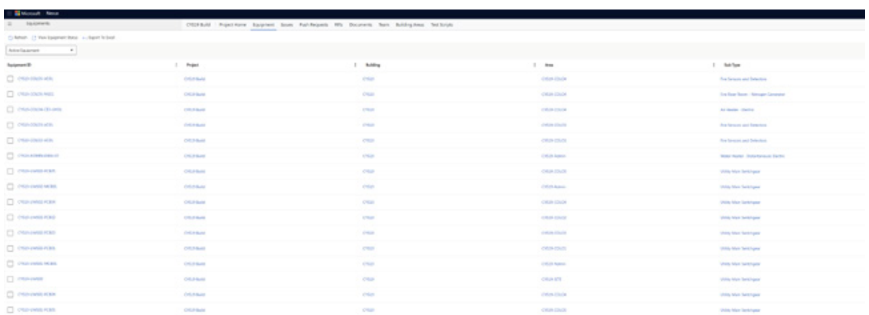
## FOK EQUIPMENT LIST

The FOK equipment list defines the scope of mockup requirements and commissioning expectations. This list includes every piece of electrical equipment that must be inspected and commissioned.

Contents:

- Equipment type (UPS, MSB, ATS, AHU, etc.)
- Source (OFCI or EC/GC-furnished)
- Vendor and point-of-contact information
- Associated checklist or inspection template required
- Identification of CYT vs. permanent power

This list is dynamic. It must be created early - ideally prior to the Electrical Partnering Meeting - and kept current throughout the project. Any changes in equipment type, vendor, or delivery timeline must be reflected promptly.



*Sample FOK Equipment List*

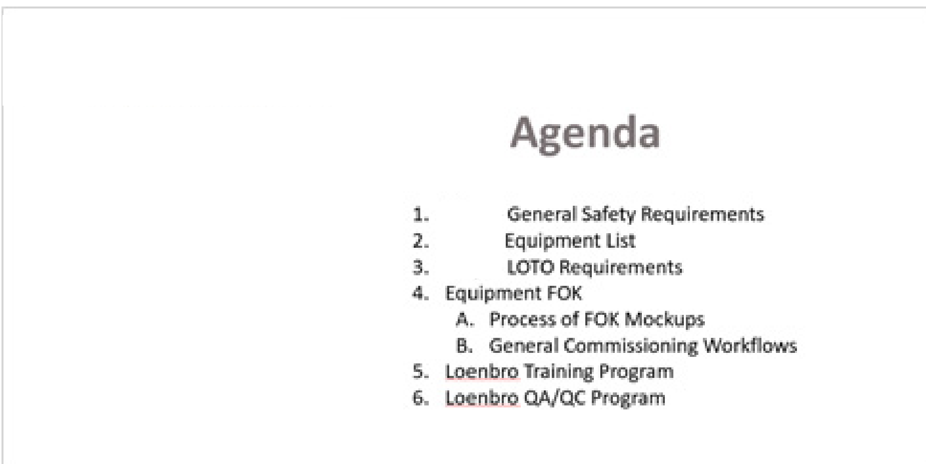
## ELECTRICAL PARTNERING MEETING

This in-depth meeting is one of the most important steps in aligning electrical scope across teams. Held shortly after EC onboarding, it facilitates agreement on safety expectations, QA/QC execution, tracking tools, and installation sequences.

Topics Include:

- EC's Site-Specific Safety Plan and LOTO Program review
- QA/QC program structure, including foreman vs. dedicated QC staff
- EC's training and certification program for installation crews
- Review of initial equipment list and responsible parties for finalization
- Selection of software/tracking tools (e.g., QR code tags, sticker systems)
- Workflow status milestones, such as:
- Material received
- Equipment installed
- Vendor checks complete
- FOD inspections complete
- Ready to energize
- Schedule of FOK inspections for each equipment type
- Review of constructability risks and design clashes

This meeting must result in distributed minutes with signatures confirming consensus. It also serves as the trigger for initiating FOK inspections.


*Sample Meeting Agenda*

## MOCKUP KICK-OFF MEETING AGENDA

Before each mockup/FOK inspection, a kickoff meeting is conducted with all necessary stakeholders to set expectations, define roles, and ensure that the mockup is ready.

Meeting Scope:

- Identify what will be inspected and where (preferably final location)
- Confirm equipment is complete and in final install state
- Review all pre-energization and vendor-specific checklists
- Define any custom inspection criteria
- Assign roles for documenting, photographing, and sign-off
- Mockup inspections must validate that:
  - All lugs are torqued, labeled, and documented
  - No FOD or installation debris is present
  - Proper hardware (Belleville washers, bolts, grommets) is used
  - Cleanliness, labeling, and routing match approved examples
  - All lessons learned are carried forward to production

## MOCKUP SIGNATORY DOCUMENTS

Mockup Signatory Documents ensure accountability and shared ownership of installation quality. At the conclusion of each FOK inspection, representatives from all attending parties must review, agree to, and sign off on:

- Final installation condition
- Verified checklist completion
- Reviewed and resolved issues or potential RFIs
- Updated workflows and expectations for future work

This documentation is uploaded to the project SharePoint and used as a training reference for the installation teams during production.

## SAMPLE MOCKUP /FIRST OF KIND ELECTRICAL EQUIPMENT INSPECTION

| | |
|---|---|
| **JOB NAME & NUMBER:**<br>LYH10 10107696 | **TYPE OF EQUIPMENT:**<br>Air Handler LYH10-COLO1-CE1-AHU02 |
| **MANUFACTURER OF EQUIPMENT:**<br>Silent Air | **DATE OF INSPECTION:**<br>05/8/2025 |
| **EQUIPMENT SERIAL #:**<br>0224-4459-27 | **CONTRACTOR PERFORMING WORK:**<br>ACI, MC Dean, Silent Air, Rosendin, Siemans |

***IDENTIFY ALL PARTIES PRESENT AT INPSECTION***

**ATTENDANCE**

| | | |
|---|---|---|
| Name: Chase Campbell | Company: Clayco | Contact: 314-924-0002 |
| Name: Brandon Stamm | Company: Clayco | Contact: |
| Name: Kristen Tucker | Company: ACI | Contact: |
| Name: Keith | Company: Silent Aire | Contact: |
| Name: Keith Morrison | Company: CAI | Contact: |
| Name: Caston Baillargeon | Company: CAI | Contact: |
| Name: Zack Hawk | Company: MC Dean | Contact: |
| Name: | Company: | Contact: |

*Sample Signatory Sheet*

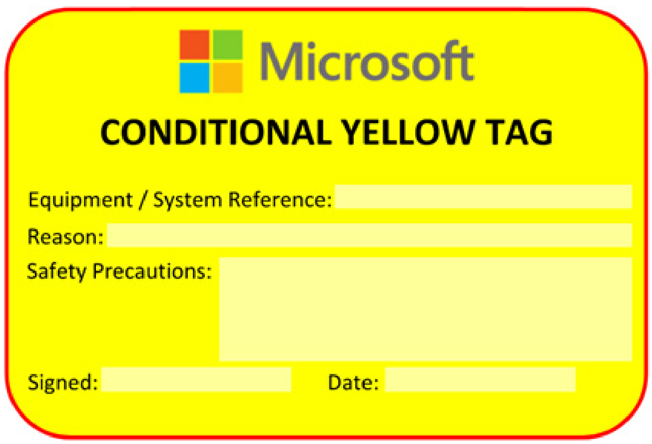
# CYT (Conditional Yellow Tag)

*The CYT process supports safe* **temporary energization** *of critical equipment for commissioning or functional testing before full power-up is authorized. CYT tagging identifies equipment that is temporarily energized under specific, controlled conditions using construction power.*

**Requirements**:

- Labeled connection points (temporary vs. permanent)
- Detailed CYT script
- CYT-specific MOP approved by GC, EC, and vendor
- CYT tags displayed at point of use

The CYT ensures temporary energization does not result in safety incidents, equipment damage, or process confusion.




*Sample CYT tag or field photo*

### CYT SCRIPT TEMPLATE

The CYT Script outlines detailed, step-by-step energization steps to be executed under temporary power conditions. It is customized for each equipment type and must include:

- Equipment model and location
- Fuse removal and restoration procedures
- Verification methods (lockout points, visual confirmations)
- Field testing and isolation steps
- Names and roles of involved personnel

This script is signed off by the EC, vendor, and QA/SME before work begins.

### CYT MOP TEMPLATE

The CYT Method of Procedure (MOP) details the sequence of activities to temporarily energize equipment for commissioning and revert it safely back to its de-energized state before final integration. It includes:

- Step-by-step energization and de-energization activities
- Lockout/tagout responsibilities and coordination points
- Verifications for equipment grounding and labeling
- Safety precautions and rescue plans

A completed CYT MOP must be uploaded to the project database and linked to the equipment tracking system.

### CYT EQUIPMENT SPECIFIC POWER SOURCE

For every piece of equipment with a Conditional Yellow Tag, a specific power source diagram must be created and approved. These diagrams show:

- Origin of temporary construction power
- Routing through panels, transformers, and breakers
- Clearly labeled connections and terminals in the field

This ensures that energized circuits are easily identifiable, and that the reversion to permanent power is methodical and safe.

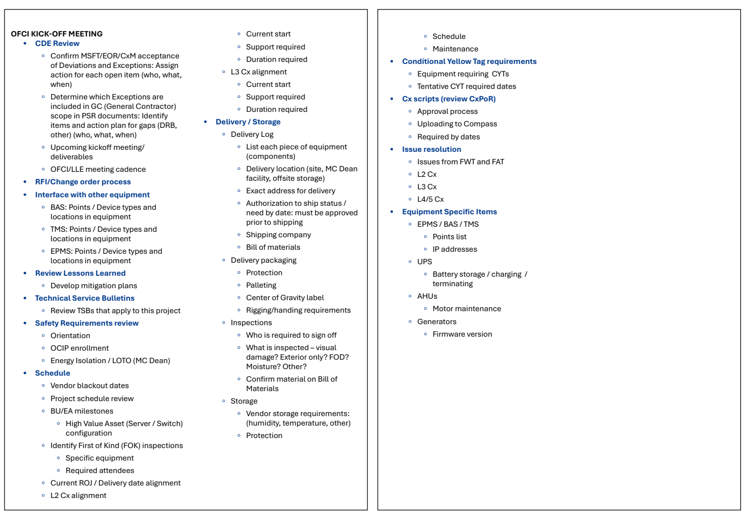
# Equipment Vendor

### EQUIPMENT VENDOR KICK-OFF MEETING TEMPLATE

This meeting establishes shared expectations with vendors and vendor agreement holders (the client in cases of OFCI) and defines project-specific requirements for:

- Delivery timelines and milestone alignments
- Receiving inspection criteria (shock tags, packaging integrity)
- Storage conditions (humidity, temperature, elevation)
- Documentation required for Factory Acceptance Test (FAT), Site Acceptance Test (SAT ), torque logs, etc.
- Commissioning support commitments (on-call vs. onsite)
- Meeting minutes must be signed and saved in the vendor coordination record.

***See Exhibit E Equipment Vendor Kickoff Meeting Template***



**OFCI KICK-OFF MEETING**

- **CDE Review**
  - Confirm MSFT/EOR/CxM acceptance of Deviations and Exceptions: Assign action for each open item (who, what, when)
  - Determine which Exceptions are included in GC (General Contractor) scope in PSR documents: Identify items and action plan for gaps (DRB, other) (who, what, when)
  - Upcoming kickoff meeting/ deliverables
  - OFCI/LLE meeting cadence
- **RFI/Change order process**
- **Interface with other equipment**
  - BAS: Points / Device types and locations in equipment
  - TMS: Points / Device types and locations in equipment
  - EPMS: Points / Device types and locations in equipment
- **Review Lessons Learned**
  - Develop mitigation plans
- **Technical Service Bulletins**
  - Review TSBs that apply to this project
- **Safety Requirements review**
  - Orientation
  - OCIP enrollment
  - Energy Isolation / LOTO (MC Dean)
- **Schedule**
  - Vendor blackout dates
  - Project schedule review
  - BU/EA milestones
    - High Value Asset (Server / Switch) configuration
  - Identify First of Kind (FOK) inspections
    - Specific equipment
    - Required attendees
  - Current ROJ / Delivery date alignment
  - L2 Cx alignment
    - Current start
    - Support required
    - Duration required
  - L3 Cx alignment
    - Current start
    - Support required
    - Duration required
- **Delivery / Storage**
  - Delivery Log
    - List each piece of equipment (components)
    - Delivery location (site, MC Dean facility, offsite storage)
    - Exact address for delivery
    - Authorization to ship status / need by date: must be approved prior to shipping
    - Shipping company
    - Bill of materials
  - Delivery packaging
    - Protection
    - Palleting
    - Center of Gravity label
    - Rigging/handing requirements
  - Inspections
    - Who is required to sign off
    - What is inspected – visual damage? Exterior only? FOD? Moisture? Other?
    - Confirm material on Bill of Materials
  - Storage
    - Vendor storage requirements: (humidity, temperature, other)
    - Protection
    - Schedule
    - Maintenance
- **Conditional Yellow Tag requirements**
  - Equipment requiring CYTs
  - Tentative CYT required dates
- **Cx scripts (review CxPoR)**
  - Approval process
  - Uploading to Compass
  - Required by dates
- **Issue resolution**
  - Issues from FWT and FAT
  - L2 Cx
  - L3 Cx
  - L4/5 Cx
- **Equipment Specific Items**
  - EPMS / BAS / TMS
    - Points list
    - IP addresses
  - UPS
    - Battery storage / charging / terminating
  - AHUs
    - Motor maintenance
  - Generators
    - Firmware version

*Sample Equipment Vendor Kick-Off Meeting Agenda*

### SHOCK DETECTION ON UPS PACKAGING FOR SHIPPING

Shock detection devices are used on high-value electrical equipment - especially UPS systems - to monitor for potential damage during transit. Shock sensors are affixed to shipping crates and must be reviewed on receipt.

Inspection Actions:

- Visual check of sensors on all four sides
- Photograph and log any triggered indicators
- Cross-reference against Bill of Lading
- Escalate to vendor for evaluation if damage is suspected

This protocol reduces the chance of energizing compromised equipment and ensures vendor accountability.


*Sample shock sensor photo*

# Equipment Vendor Engagement Program

***The objective of effective equipment vendor engagement is to ensure that all project equipment meets the highest safety and compliance standards, especially regarding arc flash protection. By establishing strong partnerships with vendors, we will enhance reliability and performance while promoting transparency. This initiative-taking approach will mitigate risks early, ensuring safe, efficient, and sustainable project delivery.***

**#1: QUALITY CONTROL PROGRAM FOR MANUFACTURERS**

A Quality Control (QC) Checklist Program is designed to be completed by equipment manufacturers prior to shipping equipment. Its purpose is to ensure that every unit delivered to a project site meets the highest standards of safety, reliability, and compliance with industry codes. By requiring manufacturers to document inspections, testing, and protective measures before shipment, this program reduces the risk of equipment failures, rework, and costly delays.

**IMPLEMENTATION**

### 1. QC Program Acceptance Mandate

- All vendors must operate under an established Quality Control (QC) program.
- Vendors shall demonstrate that each piece of equipment has successfully passed their internal QC process prior to shipment.

### 2. Minimum QC Checklist

- At a minimum, vendor QC checklists shall include verification of the following:
    - Correct equipment labeling and identification
    - Protective covers and guards installed and secured
    - Foreign object debris (FOD) check completed (no shavings, washers, loose hardware, etc.)
    - Required documentation included (manuals, test reports, certifications)
    - Packaging suitable for transportation and environmental protection

### 3. Proof of QC Completion

- Vendors must provide formal proof that QC has been performed, which may include:
    - Signed vendor certification statement
    - Completed inspection checklist/report
    - Photo documentation of inspected equipment prior to shipment

**QC CHECKLIST FOR OUTGOING EQUIPMENT**

1. **Completed Work**
    - Any equipment not completed in-house and must be certified when completed on the project.
    - Completion waiver. Flagged
    - Place into A2S as a notification.
2. **Pre-Inspection Preparation**
    - Work area cleaned, tool accountability, FOD control zones marked.
3. **Equipment Inspection**
    - Visual inspection for loose hardware, debris, fasteners torqued, covers sealed, capped.

4. **Packing & Handling**
   - Cleaned, inspected for integrity, protective wraps, no loose items, labels, and documentation.
5. **Tool & Material Accountability**
   - Tool check-in/check-out log, no authorized items in or around equipment.
6. **Final QC Sign-Off**
   - FOD checklist completed and signed by inspector, supervisor review, equipment tagged.
   - Optional: UV light for hard to see areas, barcode tracking, random FOD audits.

### #2: Equipment Delivery Inspection Form "Hand-Off"

A joint inspection between the vendor and the installer would be conducted to confirm that the equipment has been delivered in good condition and in accordance with all specified requirements. Upon receipt, the equipment would be inspected to ensure it has been adequately protected against weather, rodents, and foreign object debris. Clearly defining ownership at each stage of the process promotes accountability, reduces rework, and enhances both equipment reliability and overall job site safety.

### Implementation

### Required Individuals

1. Installer (responsible for installation readiness and verification)
2. Vendor representative (responsible for equipment condition and documentation)
3. Optional/Recommended: Project Superintendent or Safety Manager (oversight/validation)

### Standardized Equipment Receipt Form

1. Vendor provides acknowledgment of condition and readiness for installation
2. Installer signs off on receipt and inspection completion
3. Photos and signed checklist stored in project records
4. Any deficiencies documented, reported immediately, and corrective actions assigned before installation proceeds

### Inspection Location Parameters

- Packing: Re-packing and verification.
- Storage: Protection requirements such as weatherproofing, rodent protection, FOD control.
- Accessibility: Must allow safe unloading, staging, and handling of equipment.
- Surface Conditions: Level, stable ground/pad capable of supporting equipment weight.
- Lighting: Adequate natural or artificial lighting for detailed inspection.
- Environmental Protection: Overhead cover or temporary shelter for inspections during inclement weather.
- Tools & Resources: Flashlight, camera/tablet, standardized inspection checklist, PPE.
- Security: Controlled access to prevent unauthorized handling or movement of equipment prior to inspection completion.
- Waste/FOD Control: Containers available for debris, packaging, and removed FOD.
- Documentation Station: Workspace for checklists, signatures, and photo documentation.

### Inspection Requirements

Inspection should be conducted jointly at time of receipt and prior to hand-off. Tools and methods include:

- **Flashlight** – to check internal/external components, labels, and connections
- **Camera/Tablet** – to document condition, labels, and serial numbers for recordkeeping
- **Checklist Items** (examples):
  - Equipment free from visible damage (dents, cracks, corrosion, defects)
  - Nameplates, labels, and warning signage are intact and legible
  - Protective covers, guards, and barriers are in place
  - Breakers, switches, or disconnects in correct shipping/receiving position
  - Internal compartments clean, dry, and free of debris
  - Lugs and terminals free from damage and properly protected
  - Accessories and components accounted for (per packing list)
  - Arc flash and electrical hazard labeling correct and compliant
  - Documentation/manuals provided with delivery

## #3: Vendor Safety & Collaboration

Joint safety planning ensures consistency and accountability across project teams while reducing risks of complacency and miscommunication. Site personnel must be trained on specific hazards, and last-minute changes should be managed carefully. Clearly defined roles and responsibilities foster alignment among stakeholders. Without coordinated planning, communication can break down, causing delays and procedural disconnects. A structured safety planning process enhances collaboration and improves project execution safety and reliability.

### Implementation

- MSFT/Vendor recommended to train vendor team members prior to arrival on project.
- All craft workers are required attend the site-specific orientation and follow the require policies and procedures.
- Prior to energization, Vendors are required to participate in the project specific Electrical Energization Plan KO meeting and review the document.

### Compliance with Safety Standards

- All vendors and contractors must comply with safety standards before working onsite or delivering equipment.
- A documented Activity Hazard Analysis (AHA) or equivalent “path of action” must be submitted to demonstrate how work will be conducted safely.
- Vendors are required to acknowledge Vendor Energization Program deliverable checklist. This checklist is distributed during the KO meeting and shared with the respective Project Managers for visibility and tracking.

### Coordination Calls

- Each coordination call must include a competent representative with knowledge of site logistics, planning, and safety requirements.
- A field representative is required to be present on commissioning (CX) and daily planning calls.

### Confined Space & High-Risk Activities (HRAs)

- Confined space requirements must be reviewed with all vendors before work begins.
- High-Risk Activity (HRA) expectations and responsibilities must be clearly identified, communicated, and incorporated into planning.

### Tiered Trade Partner Notification

- Vendors must provide advance notification if a tiered trade partner is scheduled to be onsite.
- Tiered trade partners are subject to the same prequalification and compliance requirements.

# Exhibits
# Table of Contents

# Exhibit A
# DAILY FOD CHECKLIST TEMPLATE

## DAILY 'FOD' FOREIGN OBJECT DEBRIS CHECKLIST

### Section 1 – General Information

**Contractor Name:**____________________________________________________________

**Location of Work / Name of Equipment:**______________________________________________

**Date and Time of Work:**______________________________________________________

**Description of Work:** ________________________________________________________

________________________________________________________________________

**Name of Worker(s):** ____________________ ____________________ ____________________

____________________ ____________________ ____________________

____________________ ____________________ ____________________

### Section 2 – Checklist

***Training***

1. Confirm that all crew members are trained and aware of FOD hazards and reporting procedures. YES_____ NO_____

***Debris Free Zone***

2. Ensure the work environment is clean and free from debris and dust that could contaminate equipment. YES_____ NO_____

***Equipment Covers / Temporary Covers are Present***

3. Ensure all panels, distributions, or any electrical equipment have covers in place prior to performing scope. YES_____ NO_____

***If covers are not present, stop work and notify your GC and Energy Marshal to identify the last crew that was working on that equipment.***

***Removal of Covers / Temporary Covers***

4. Ensure all covers, bolts, screws and miscellaneous hardware is stored and secured away from the equipment. YES_____ NO_____

***Securing Loose Items***

5. Verify all loose materials, tools, hardware and equipment is secured and protected from becoming FOD. YES_____ NO_____

***Lighting***

6. Verify that there is adequate lighting to identify and remove FOD. YES_____ NO_____

***Modifications***

7. Verify protection from FOD is installed during drilling, taping, cutting or knocking out any piece of equipment. YES_____ NO_____

***Provide temporary protection if equipment is left unintended during lunch, breaks or overnight if permanent covers are not available.***

***Notify your Supervisor if and materials or hardware is dropped within the equipment and cannot be located or retrieved.***

### Section 3 – Final Inspection (End of Shift)

8. Verify all tools, hardware, materials etc. has been removed from equipment & accounted for prior to covering YES_____ NO_____
9. Ensure all equipment has been cleaned and vacuumed prior to installing covers YES_____ NO_____
10. Verify all equipment has been closed up/protected from FOD entry on fronts, sides, and top at the end of shift YES_____ NO_____
11. Verify that all areas of work have been broom swept prior to leaving at the end of shift YES_____ NO_____

**FOD Manager Name:**__________________________ **FOD Manager Signature:**______________________________

**Date:**________________________ **Time:**________________________

**Comments:**______________________________________________________________________

# Exhibit B
# FOD PREVENTION PLAN TEMPLATE

**FOD Prevention Plan**

**Project:** ______________________________________________

**Date:** ______________________________________________

**Controlling Contractor:** ______________________________________________

**Define FOD Prevention RACI Matrix for this project:**

| | R = Responsible; A = Accountable; C = Consult; I = Inform | | | | |
|---|---|---|---|---|---|
| **Description** | **GC safety** | **GC Supt.** | **EC** | **Vendor** | **Owner Safety** |
| Define FOD Prevention Areas | R | A | C | I | I |
| Review equipment handling plans with vendor | A | A | R | C | I |
| Task Planning / FOD Mitigation Techniques | | | | | |
| Housekeeping Plan (establish before GRs?) | C | R | C | I | I |
| Housekeeping inspections / audits? | I | R | R | | I |
| Equipment protection inspections | I | A | R | R | I |
| Employee training plan development | R | C | C | C | I |
| Employee FOD specific training | R | C | R | R | I |
| FOK (First of Kind Inspection) plan | C | R | C | C | I |

1. **Define FOD Prevention Areas**
   - **FOD Awareness Areas**: General areas where equipment maintenance and modification occur. Ensure these areas are free from potential FOD entrapment.
   - **FOD Awareness Areas on this project:**

     ____________________
     ____________________
     ____________________
     ____________________

   - **FOD Control Areas:** Areas where gear assembly or modification processes are ongoing. Implement strict controls to prevent FOD migration.
   - **FOD Control Areas on this project:**

     ____________________
     ____________________
     ____________________
     ____________________

   - **FOD Critical Areas:** Areas requiring the highest level of preventative measures, such as withing equipment rooms and within the gear itself.
   - **FOD Critical Areas on this project:**

     ____________________
     ____________________
     ____________________
     ____________________

2. **Equipment handling measures**
   - Packaging, handling, storage and transportation such that no FOD is introduced into the equipment as it is being brought into the building as a new delivery.
   - Ensure transportation of equipment onsite does not introduce any FOD debris within the equipment.
   - **Specific equipment handling measures on this project:**

     ____________________
     ____________________
     ____________________
     ____________________

3. **Implement Housekeeping Practices Onsite**
    - **Controlled entrances and pathways into the building**
        - Paved or clean stone pathways into the building – as soon as SOG is complete
        - Walk-off mats / Skudo mats at building entrances
    - **Specific controlled entrances and pathways on this project:**

      ________________________________________
      ________________________________________
      ________________________________________
      ________________________________________

    - **Organized Work Areas**
        - Keep aisles, work benches, toolboxes, carts, and outside areas clean and organized
        - Tool Management: Keep tools organized and stored in designated areas when not in use
        - Material Storage: Store materials and parts in specific locations to avoid clutter and potential FOD
        - Labeling: Clearly label storage areas and containers to ensure everything is in its proper place
        - Debris-Free Zones: Ensure the work environment is clean and free from debris that could contaminate equipment
            - Reduce dust while installation of equipment is ongoing. Drywall cutting should be moved outside or away from the equipment areas
        - Rodent mitigation plan should be on all projects to reduce the chance of damage or risk
        - Implement 6S
            - Sort : Identify and remove unnecessary items from the workplace to reduce clutter and potential hazards.
            - Set in Order: Organize and label necessary items in a logical and accessible manner to ensure efficient workflows.
            - Shine: Regularly clean and maintain the workplace to keep it in pristine condition.
            - Standardize: Establish consistent work processes, responsibilities, and visual cues to maintain the 6S system.
            - Sustain: Maintain the system over the long term through ongoing training, audits, and continuous improvement.
            - Safety: Identify hazards and set preventive controls to keep workers safe during operations and ensure the work environment meets required safety standards
        - **Specific organized work areas / measures on this project:**

          ________________________________________
          ________________________________________
          ________________________________________
          ________________________________________

    - **Clean As You Go**
        - Continuously remove debris during manufacturing, fabrication, modification, operations, or maintenance to ensure the product is FOD-free
        - Immediate Disposal: Dispose of any foreign objects or debris immediately to prevent accumulation
        - Continuous cleaning in COLOs
        - **Specific clean as you go measures on this project:**

          ________________________________________
          ________________________________________
          ________________________________________
          ________________________________________

- **Designated Cleaning Times**
  - Scheduled Cleanups: Set specific times for thorough cleaning of work areas, such as at the end of each shift
  - Team Efforts: Encourage team-based cleaning efforts to ensure all areas are covered
  - **Specific cleaning times on this project:**

- **Use of Proper Cleaning Tools**
  - Appropriate Tools: Use the right tools and equipment for cleaning to ensure thorough removal of debris
  - Tool Maintenance: Regularly maintain and replace cleaning tools to ensure they are effective
  - **Specific cleaning tools used on this project:**

- **Metal Shaving prevention**
  - No cutting metal studs in the building – use shears, cut outside the building, isolated/enclosed cutting station inside the building
  - No cutting Unistrut in the building - use shears, cut outside the building, isolated/enclosed cutting station inside the building
  - UPS ductwork attachment details / electrical top hat attachment details that do not require access to the inside of the equipment or result in metal shavings from screws getting into the equipment.
  - **Specific metal shaving prevention measures on this project:**

- **Use of Physical Barriers and Protective Covers**
  - Barriers and Covers: Use protective covers and barriers to shield equipment from potential FOD sources
  - Seal Openings: Ensure all openings and access points are sealed when not in use to prevent debris from entering
  - Vents/openings covered
  - Protect transformer coils (e.g. PDUs)
  - Designated Storage Areas: Ensure tools, parts, and materials are stored in designated areas away from equipment
  - **Specific use of physical barriers and protective covers on this project:**

4. **Conduct Regular Inspections**
    - Daily Checks: Conduct daily inspections of work areas to identify and remove any foreign objects
    - Pre-energization QC checks shall be performed by a qualified individual and documented.
    - Inspection Logs: Maintain logs of inspections to track housekeeping efforts and identify recurring issues
    - **Specific regular inspections on this project:**

5. **Immediate Action on Detection**
    - Prompt Repairs: Address any signs of FOD immediately to prevent further damage
    - Professional Inspection: Seek professional inspection and diagnosis if metal shavings are detected to determine the cause and extent of the damage
    - **Specific action on detection on this project:**

6. **Documentation and Reporting**
    - Document all procedures, policies, and personnel responsible for FOD prevention outside of the Cx POR.
    - Reporting Mechanisms: Implement mechanisms for reporting FOD inspections, incidents and tracking corrective actions
    - **Specific documentation and reporting on this project:**

7. **Employee Training and Awareness**
    - Training Programs: Implement training programs to educate employees on the importance of housekeeping and FOD prevention
    - Awareness Campaigns: Conduct regular awareness campaigns to reinforce good housekeeping practices
    - **Specific employee training and awareness on this project:**

# Exhibit C
# PRE-ENERGIZATION FOD INSPECTION TEMPLATE

## FOREIGN OBJECT DEBRIS (FOD) INSPECTION FORM

PROJECT: ______________________________

INSPECTION DATE: ______________________________

EQUIPMENT TAG/LABEL: ______________________________

INSPECTION COMPANY: ______________________________

INSPECTED BY: ______________________________

| | |
|---|---|
| **WORK AREA ASSESSMENT:** | |
| | The work area is free of foreign objects (FOD), including loose parts, tools, materials, liquids, and other potential contaminants. |
| | The designated workspace for equipment installation has been cordoned off. |
| | Protection has been put in place when penetrating the equipment, such as magnet bags. |
| **TOOL ACCOUNTABILITY:** | |
| | All tools have been inspected for damage, wear, or loose parts and are in a safe and serviceable condition. |
| | All tools and materials have been accounted for, and there are no tools or materials unattended in the workspace when unoccupied. |
| | All tools and miscellaneous materials not currently in use during the installation are stored properly outside the workspace. |
| | |
| **EQUIPMENT INSPECTION:** | |
| | All connectors and terminals have been visually inspected for signs of damage, corrosion, or loose connections. |
| | All wiring has been visually inspected for signs of damage, fraying, or loose insulation. |
| | All equipment components have been inspected for any signs of FOD, such as loose parts or debris. |
| | The equipment floor has been inspected for signs of any FOD, such as loose parts or debris. |
| | All removed panels have been reinstalled. |
| | All FOD protection measures, such as vents and opening temporary covers, have been installed and are properly secured. |
| | Equipment has been inspected for dents, scratches, or other markings indicating potential impact damage. |
| | |
| **REPORTING:** | |
| | Any issues found during this inspection have been reported to the General Contractor. |
| | |

| | |
|---|---|
| ISSUE FOUND: | |
| REPORTED TO: | |
| CORRECTIVE ACTION: (IF ANY) | |

**OTHER ITEMS / REMARKS:**

# Exhibit D
# ROOM READINESS CHECKLIST FOR ENERGIZATION TEMPLATE

**General Contractor** Name Here
**Project** Name Here

| | Room Readiness Checklist before Energization | Date Verified |
|---|---|---|
| | **Room name / #** | |
| 1 | Doors locked (except primary access) | |
| 2 | Permanent fire extinguishers in place | |
| 3 | Room clean | |
| | a. Electrical gear | |
| | b. Mechanical equipment | |
| | c. Overhead MEP | |
| | d. Architectural surfaces | |
| | e. No stored materials | |
| 4 | Fire safing of penetrations | |
| 5 | Electrical energization cart in place | |
| | a. AED | |
| | b. Shepards hook | |
| | c. Halatron Fire extinguisher | |
| | d. Evacuation horn | |
| | e. Emergency eye wash | |
| | f. Battery light | |
| | g. First aid kit | |
| | f. Fire blanket | |
| 6 | Signage in place: (On / Outside of Primary Access door) | |
| | a. Room name / # | |
| | b. Permit sleeve (on wall next to door) | |
| | c. Danger Energized Room – Permit access only | |
| | d. Electrical Energy Isolation Sign (Energy Marshals) | |
| | e. List of people approved on the permit | |
| | | |
| | | |
| | Verification By: | |

# Exhibit E
# EQUIPMENT VENDOR KICK-OFF MEETING TEMPLATE



**OFCI KICK-OFF MEETING**

- **CDE Review**
  - Confirm MSFT/EOR/CxM acceptance of Deviations and Exceptions: Assign action for each open item (who, what, when)
  - Determine which Exceptions are included in GC (General Contractor) scope in PSR documents: Identify items and action plan for gaps (DRB, other) (who, what, when)
  - Upcoming kickoff meeting/deliverables
  - OFCI/LLE meeting cadence
- **RFI/Change order process**
- **Interface with other equipment**
  - BAS: Points / Device types and locations in equipment
  - TMS: Points / Device types and locations in equipment
  - EPMS: Points / Device types and locations in equipment
- **Review Lessons Learned**
  - Develop mitigation plans
- **Technical Service Bulletins**
  - Review TSBs that apply to this project
- **Safety Requirements review**
  - Orientation
  - OCIP enrollment
  - Energy Isolation / LOTO (MC Dean)
- **Schedule**
  - Vendor blackout dates
  - Project schedule review
  - BU/EA milestones
    - High Value Asset (Server / Switch) configuration
  - Identify First of Kind (FOK) inspections
    - Specific equipment
    - Required attendees
  - Current ROJ / Delivery date alignment
  - L2 Cx alignment
    - Current start
    - Support required
    - Duration required
  - L3 Cx alignment
    - Current start
    - Support required
    - Duration required
- **Delivery / Storage**
  - Delivery Log
    - List each piece of equipment (components)
    - Delivery location (site, MC Dean facility, offsite storage)
    - Exact address for delivery
    - Authorization to ship status / need by date: must be approved prior to shipping
    - Shipping company
    - Bill of materials
  - Delivery packaging
    - Protection
    - Palleting
    - Center of Gravity label
    - Rigging/handing requirements
  - Inspections
    - Who is required to sign off
    - What is inspected – visual damage? Exterior only? FOD? Moisture? Other?
    - Confirm material on Bill of Materials
  - Storage
    - Vendor storage requirements: (humidity, temperature, other)
    - Protection

        - Schedule
        - Maintenance
- **Conditional Yellow Tag requirements**
    - Equipment requiring CYTs
    - Tentative CYT required dates
- **Cx scripts (review CxPoR)**
    - Approval process
    - Uploading to Compass
    - Required by dates
- **Issue resolution**
    - Issues from FWT and FAT
    - L2 Cx
    - L3 Cx
    - L4/5 Cx
- **Equipment Specific Items**
    - EPMS / BAS / TMS
        - Points list
        - IP addresses
    - UPS
        - Battery storage / charging / terminating
    - AHUs
        - Motor maintenance
    - Generators
        - Firmware version

First of Kind (FOK)

## Field Inspection Checklist

| OBSERVATION DETAILS | | | | | |
|---|---|---|---|---|---|
| **Project Name and #:** | **Location/Area:** | **Observation Type:** | **Observation Date:** | **Reported By:** | **Reviewed With:** |
| NAME | COLO# / Cell# | FOK installation | Date | Name / Company | GC: Name / Company<br>EC: Name / Company<br>Vendor: Name / Company<br>EOR: Name / Company<br>MSFT: Name |

| **Equipment Name** | **GC** | **EC** | **Vendor** | **EOR** | **MSFT** | **Date** | **Comments/Observations:** |
|---|---|---|---|---|---|---|---|
| Terminations per specifications and equipment submittals | | | | | | | |
| Conductor supports per equipment shop drawings and reviewed by vendor | | | | | | | |
| Attachments to equipment (chimneys, side cars, top hats) reviewed by vendor, EOR | | | | | | | |
| EPMS / BAS biscuit location | | | | | | | |
| No signs of water infiltration ** if signs of water infiltration the source must be identified | | | | | | | |
| Labels installed – correct type and locations | | | | | | | |
| No signs of FOD / metal shavings ** if metal shavings are found the source must be identified | | | | | | | |
| Verify no physical damage | | | | | | | |
| Confirm proper grounding | | | | | | | |
| Confirm proper anchoring | | | | | | | |
| Note: Add equipment specific items based on submittals, TSBs issued, etc. that need to be reviewed during FOK inspection. | | | | | | | |
| | | | | | | | |
| | | | | | | | |
| | | | | | | | |

**Page 1**

First of Kind (FOK)

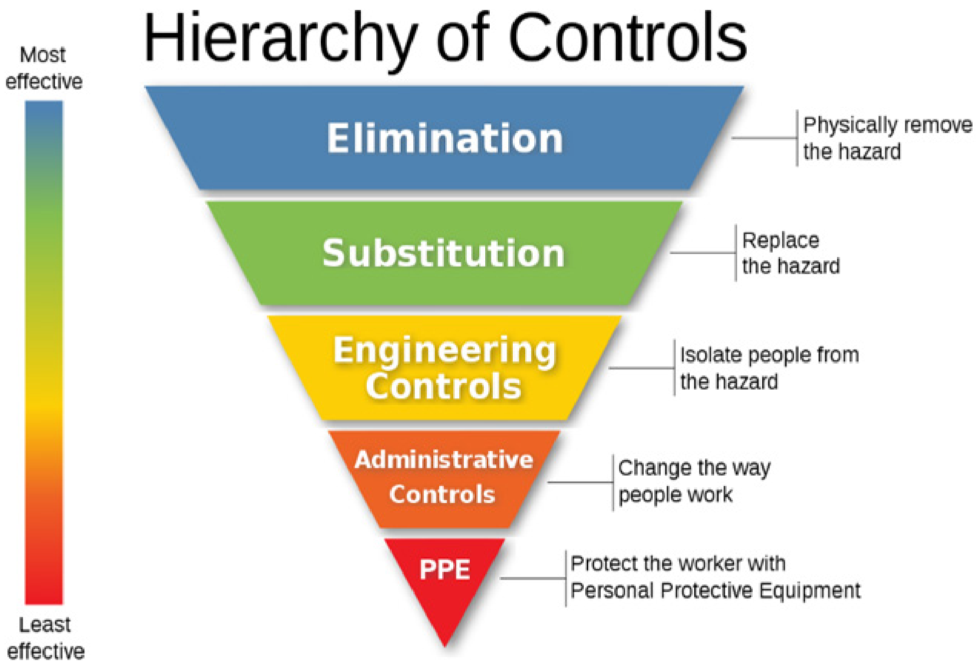
# Field Inspection Checklist

| Equipment Name | GC | EC | Vendor | EOR | MSFT | Date | Comments/Observations: |
|---|---|---|---|---|---|---|---|
| | | | | | | | |

Page 2

# Exhibit G
# HIERARCHY OF CONTROLS (GRAPHIC) AND DEFINITIONS

Following the Hierarchy of Controls is an especially useful basis for this.




1. **Elimination** is the primary and most effective method to reduce the potential for an Arc Flash or Arc Blast incident occurring by isolating the equipment and removing the energy including and stored or charged energy being discharged. It should be remembered that the act of performing the isolation involves the risk of an Arc Flash incident involving the authorised personnel performing the isolation.

2. **Substitution / Replacing** the hazard by reducing protection settings to reduce incident levels or by replacing equipment with equipment which offers greater Arc Flash protection such as equipment with active monitoring of potential incidents or equipment which offers passive protection by being designed and manufactured to contain and incidents and allow any Arc flash or arc Blast to escape through the top or rear of the equipment, directing it away from the operator. Equipment is also available which uses a vacuum or arc quenching medium reduce the likelihood of an arc occurring or becoming significant. In LV systems it is not common for these circuit breakers to be used rather circuit breakers which open and close in air.

3. **Engineering Controls** It is accepted that when certain tasks are being performed there is an increased likelihood of conductors opening or closing which may increase the potential for an Arc Flash or arc Blast incident to occur such as when circuit breakers are opening and closing or when systems are under load during testing. It is best practice that areas are vacated at such times to ensure unfamiliar staff who may not be aware of the hazard or have employed any mitigation, and it is critical that the permit system limits and controls access at these times. For personnel operating the equipment, remote switching and remote racking of circuit breakers can be employed to remove them from the area effected by the potential hazard however, in certain cases, equipment design does not allow for this method of control.

4. **Administrative Controls**, by increasing awareness and the ability of staff to identify risks and potential mitigation measures through training, we increase the likelihood of their safety. Personnel who operate equipment require further training and must have any equipment or personal protective equipment provided to them, this in turn must be monitored and maintained. Only trained and authorised personnel who are familiar, operate equipment, only trained and authorised personnel approve permits for work in energised areas, the equipment including PPE identified as mitigation measures are available, switching schedules and system boundaries are utilised, proving systems dead is enforced and practiced, skilled electrical personnel and electricians only work on fully isolated systems, equipment is deep cleaned prior to energisation and areas are dust and debris free prior to energisation. Posting warning signage, Arc Flash Warnings and using barriers are also critical in communicating potentially hazardous areas to personnel in effected areas and to prevent unintentional encroachment during commissioning or switching activities which increase the likelihood of an Arc Flash or Arc Blast incident. This is also considered in legislation which requires an employer to provide adequate information and written instruction.

5. **Personal Protective Equipment**, the last measure to consider and the least effective is the selection and wearing of PPE. PPE must be dawned and worn correctly to ensure it will function as required. When providing or provided with PPE it is essential that it is maintained correctly as per manufacturer's instructions. It is paramount that the whole body is protected equally as opposed to increased protection for the vital organs.

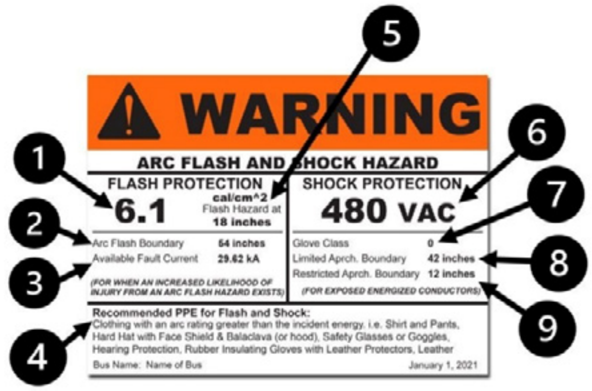
# Exhibit H
# WARNING SIGNS (GRAPHIC) & DEFINITIONS



**WARNING SIGNS**

1. **Incident Energy:** Incident energy, as indicated on the arc flash label, is the level of thermal energy that one would feel at the working distance in the event of an arc flash. This energy is measured in calories per centimetre squared. Commonly noted as cal/cm2.

2. **Arc Flash Boundary:** The Arc flash boundary is the distance from a possible arc source to where the incident energy drops to 1.2 calories per square centimetre. If a person were standing at this distance from the arc source, they could receive a second-degree burn.




3. **Available fault current:** The amount of current supplied by the electrical equipment's upstream device.

4. **Recommended PPE:** A recommended list of shock and arc flash hazard PPE that shall be worn when performing safe work on equipment that is not placed in an electrically safe condition.

5. **Working Distance:** The working distance is the distance at which the incident energy is calculated. IEEE 1584 defines the working distance as "the dimension between the possible arc point and the head and body of the worker positioned in place to perform the assigned task."

6. **Nominal System Voltage:** Nominal voltage is a value assigned for conveniently designating a voltage class, such as 230/ 480v etc.

7. **Glove Class:** Workers must be protected from shock by wearing rubber insulating gloves if their hands enter the restricted approach boundary. Workers must select gloves that have a rating for the level of voltage to which the gloves will be exposed.

8. **Limited Approach Boundary:** The limited approach boundary is designed to keep unqualified workers safe from shock hazards. Unqualified worker can only cross this boundary if he or she is continuously escorted by a qualified worker.

9. **Restricted Approach Boundary:** Only qualified persons may cross into the restricted approach boundary. Inside this boundary, accidental movement can put a part of the body or conductive tools in contact with live parts.

# Exhibit I
# LOCK AND TAG-OUT SYSTEM PROCEDURE & DEFINITIONS

**LOCK AND TAG‐OUT SYSTEM PROCEDURE**

**Definitions**

- **Energy Source:** Any source of electrical, mechanical, hydraulic, pneumatic, chemical, thermal or any other energy source.
- **Energy Isolation Device:** Is always a mechanical unit that physically blocks or interrupts the flow or release of Hazardous Energy. Equipment will be isolated when every energy source has been blocked or removed by an isolation device and whereby inadvertent re-energisation has been made not possible.

  Pushbutton, selector switches and control circuit type devices are not acceptable electrical isolating devices. Examples: Electrical- Circuit Breaker that interrupts all phases; Hydraulic- Line valve or Spade. Mechanical: Locking bolt or Brake.
- **Key Cabinet:** a secure lockable key storage cabinet. The lockout coordinator will maintain each cabinet.
- **Lockout**: the locking of an energy isolation point on a piece of equipment or system which is fitted in such a way that the piece of equipment cannot be energised.
- **Lockout Device:** a lockable mechanical device that can be placed on an energy isolation device on an equipment or system to prevent it from being re-energised. A lockout device may allow multiple lockout devices to be placed onto a single energy lockout point, e.g. hasp.
- **Lock Out Tag:** a visual aid which is attached to a lock out device to indicate that the device is locked out and which contains a legend such as 'HOLD OFF'or 'DO NOT OPERATE' and details of who has applied the lock out device, how to contact them and the isolation record number.
- **Primary Lock Out/ Tag Out:** the first lock, lockout device and tag applied to the isolation device. Primary lock out can only be carried out the Operator in charge / SystemOwner.
- **Tag Out:** the placement of a tag on an isolation point. Tag out will only be used only in conjunction with positive lock out. Tags alone are not sufficient for isolation of hazardous energy on the energy.
- **Line Breaking:** the physical non-routine breaking into a line, a vessel or equipment known or suspected to contain hazardous material. Physical breaking includes unscrewing, unbolting or cutting of screwed type, flanged, cemented, welded or other types of connections on pipelines or process equipment. It also includes the removal of tri-clover connections from utility systems known or suspected to contain hazardous material.
- **Blanking/Blinding/Spading:** the absolute closure of a pipe, line or duct by the fastening of a solid plate (example being a spectacle blind or a skillet blind) that completely covers the bore / duct diameter and that is capable of withstanding the maximum pressure of the pipe, line or duct, with no leakage given beyond the inserted plate.
- **Group Isolation:** where more than one lock is to be applied to an isolation point a Group Isolation may be undertaken using a Group Isolation Lock Out Tag Out Box or Multihasp. The Group Isolation Box or Multihasp must be clearly identified as to the reference point of isolation in particular if this is remote from the isolation point.

# Exhibit J
# ACRONYMNS

**ACMS |** Automated Commissioning Management System

**BIM |** Building Information Modeling

**CAP |** Corrective Action Plan

**CAZ |** Controlled Access Zone

**CxA |** Commissioning Authority

**CxM |** Commissioning Manager

**CxPoR |** Commissioning Plan of Record

**CYT |** Conditional Yellow Tag

**EIP |** Energy Isolation Permit

**EMOP |** Electrical Method of Procedure

**FAT |** Factory Acceptance Test

**FOD |** Foreign Objects and Debris

**FOK |** First of Kind

**FR |** Flame Resistant

**HEPA |** High Efficiency Particulate Air - a type of air filter designed to trap very small particles

**HRA |** High-Risk Activity

**IST |** Integrated Systems Test

**MEP |** Mechanical, Electrical, Plumbing

**MSB |** Main Switchboar

**PDU |** Power Distribution Unit

**PPE |** Personal Protective Equipment

**QA |** Quality Assurance

**QC |** Quality Control

**RACI |** Responsible, Accountable, Consulted, Informed

A RACI matrix is a project management tool used to clarify roles and responsibilities for tasks or deliverables within a project or process. The acronym RACI stands for:

- **R – Responsible:** The person or people who actually do the work to complete the task.
- **A – Accountable:** The person ultimately answerable for the correct and thorough completion of the task. There should only be one accountable person per task.
- **C – Consulted:** People who provide input, expertise, or feedback. These are typically two-way communications.
- **I – Informed:** People who need to be kept updated on progress or decisions. This is typically one-way communication.

**RCA |** Root Cause Analysis

**UPS |** Uninterrupted Power Supply

**SAT |** Site Acceptance Test

**THANK YOU TO OUR CONTRIBUTING EDITORS**